

Міністерство освіти і науки України

Київський національний університет імені Тараса Шевченка

Фізичний факультет, кафедра загальної фізики

ДОДАТКОВІ МАТЕРІАЛИ

до дисертації на здобуття наукового ступеня

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*«Характеризація дефектів фотоелектричних перетворювачів
на основі вольт-амперних вимірювань та глибокого навчання»*

Розділ 5: Оцінка концентрації заліза за відносними змінами фотоелектричних параметрів
з використанням алгоритмів машинного навчання

Здобувач:

Завгородній Олексій Володимирович

аспірант кафедри загальної фізики

Науковий керівник:

Оліх Олег Ярославович, д.ф.-м.н.

професор кафедри загальної фізики

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Table S5.1. Hyperparameter space for RF

Hyperparameter	Values
# estimators	100, 200, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700
max depth	10, 15, 20, 25, 30, 35, 40, 45
min samples leaf	1, 2, 3, 4, 5, 6, 7
min samples split	2, 3, 4, 5, 6, 7
bootstrap	True, False
max features	'log2', 'sqrt', 1.0, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2

Table S5.2. Hyperparameter space for GB

Hyperparameter	Values
# estimators	100, 200, 250, 300, 350, 400, 450, 500, 550, 600, 650
max depth	15, 20, 25, 30, 35, 40, 45
min samples leaf	1, 2, 3, 4, 5, 6, 7
min samples split	2, 3, 4, 5, 6, 7
learning rate	$[10^{-3}, 10^{-1}]$
max features	'log2', 'sqrt', 1.0, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2

Table S5.3. Hyperparameter space for XGB

Hyperparameter	Values
booster	gbtree, gblinear, dart
max depth*	3, 4, 5, 6, 7, 10, 15, 20
min split loss*	$[10^{-6}; 5]$
min child weight*	$[0; 15]$
subsample*	0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0
colsample by tree*	0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0
# estimators	200, 300, 400, 500, 600, 700, 800, 900
learning rate	$[10^{-5}; 1]$
L1	$[10^{-8}; 1]$
L2	$[10^{-8}; 10]$

* for all boosters except gblinear

Table S5.4. Hyperparameter space for SVR

Hyperparameter	Values
kernel	linear, poly, rbf, sigmoid
degree*	2, 3, 4, 5, 6
C0	$[0; 5]$
Tolerance	$[10^{-5}; 10^{-2}]$
C	$[10^{-2}; 15]$
Epsilon	$[10^{-3}; 1]$

* for poly kernel only

Table S5.5. Hyperparameter space for DNN

Hyperparameter	Values
hidden layers configuration*	Pipe, Trapezium, Triangle, Butterfly, Fir
# nodes for first hidden layer	5, 10, 20, 30, 50, 75, 100, 120, 150, 200, 250
# hidden layers**	5, 6, 8, 10, 12, 15
batch size	8, 16, 32, 64, 128
activation function	ReLu, sigmoid, tanh, SELU, ELU
optimizer	SGD, RMSprop, Adam, Adadelata, Adagrad, Adamax, Nadam, Ftrl
learning rate	$[10^{-5}; 10^{-2}]$
# epochs	100, 300, 400, 500, 600, 700, 1000, 1500
weight initializer	Xavier Normal, Xavier Uniform, He Normal, He Uniform, Random Normal, Random Uniform

* The configurations are shown in Fig.S2.

** For Pipe configuration only

Table S5.6. Chosen hyperparameter combinations for RF models

Model	Hyperparameter					
	# estimators	max depth	min sample split	min sample leaf	max features	bootstrap
RF_4^{AM}	200	15	3	1	1.0	True
$\text{RF}_{4:\text{PC}}^{\text{AM}}$	650	15	2	1	0.9	True
RF_4^{940}	500	30	2	1	0.9	True
$\text{RF}_{4:\text{PC}}^{940}$	200	20	2	1	0.9	True
RF_5^{AM}	600	20	2	1	0.6	False
$\text{RF}_{5:\text{PC}}^{\text{AM}}$	300	15	4	1	0.6	False
RF_5^{940}	500	35	2	1	1.0	True
$\text{RF}_{5:\text{PC}}^{940}$	700	45	2	1	0.6	True
RF_6^{AM}	500	30	2	1	0.5	False
$\text{RF}_{6:\text{PC}}^{\text{AM}}$	600	30	2	1	0.6	False
RF_6^{940}	400	30	2	1	0.5	False
$\text{RF}_{6:\text{PC}}^{940}$	500	25	2	1	0.7	False
RF_7^{AM}	600	45	2	1	0.7	False
$\text{RF}_{7:\text{PC}}^{\text{AM}}$	550	40	2	1	0.6	False
RF_7^{940}	450	40	2	1	0.7	False
$\text{RF}_{7:\text{PC}}^{940}$	450	30	2	1	0.6	False

Table S5.7. Chosen hyperparameter combinations for GB models

Model	Hyperparameter					
	# estimators	max depth	min sample split	min sample leaf	max features	learning rate
GB_4^{AM}	500	15	4	7	0.8	8.3e-03
$\text{GB}_{4:\text{PC}}^{\text{AM}}$	550	15	5	7	0.8	8.1e-03
GB_4^{940}	650	30	6	6	0.8	1.1e-02
$\text{GB}_{4:\text{PC}}^{940}$	650	45	4	7	0.7	1.3e-02
GB_5^{AM}	650	45	2	4	0.8	4.3e-02
$\text{GB}_{5:\text{PC}}^{\text{AM}}$	550	15	4	3	0.6	7.2e-03
GB_5^{940}	600	40	4	6	0.7	1.9e-02
$\text{GB}_{5:\text{PC}}^{940}$	600	40	6	6	0.5	3.8e-02
GB_6^{AM}	600	45	7	5	0.7	3.4e-02
$\text{GB}_{6:\text{PC}}^{\text{AM}}$	550	30	7	7	0.6	1.9e-02
GB_6^{940}	650	45	7	7	0.7	2.7e-02
$\text{GB}_{6:\text{PC}}^{940}$	400	35	3	7	0.9	3.5e-02
GB_7^{AM}	450	15	4	6	0.7	2.3e-02
$\text{GB}_{7:\text{PC}}^{\text{AM}}$	550	40	2	6	0.6	2.1e-02
GB_7^{940}	650	15	5	7	0.7	2.8e-02
$\text{GB}_{7:\text{PC}}^{940}$	600	40	3	5	0.6	2.7e-02

Table S5.8. Chosen hyperparameter combinations for XGB models

Model	Hyperparameter									
	booster	max depth	min split loss	min child weight	sub sample	colsample bytree	# estimators	learning rate	L1	L2
XGB ₄ ^{AM}	dart	10	5.3e-4	10.8	0.6	1	400	1.7e-2	2.8e-2	3.1e-2
XGB _{4:PC} ^{AM}	gbtree	15	1.9e-3	11.1	0.7	1	500	9.5e-3	1.8e-4	2.6e-3
XGB ₄ ⁹⁴⁰	dart	15	2.3e-5	5.2	0.7	1	700	1.1e-2	6.6e-3	5.3e-2
XGB _{4:PC} ⁹⁴⁰	gbtree	15	1.8e-6	4	0.7	1	800	8.2e-3	1.1e-4	1
XGB ₅ ^{AM}	gbtree	15	1.1e-5	1.9	0.7	1	800	7.3e-2	9.9e-4	9.8
XGB _{5:PC} ^{AM}	dart	15	1.6e-5	10.4	0.3	1	800	6.7e-3	7.0e-3	0.5
XGB ₅ ⁹⁴⁰	dart	20	1.3e-5	1.5	0.7	1	900	1.1e-2	1.4e-4	2.4
XGB _{5:PC} ⁹⁴⁰	dart	20	1.3e-4	6.9	0.6	1	900	1.1e-2	2.5e-3	1.4e-3
XGB ₆ ^{AM}	dart	20	5.5e-5	0.7	0.5	1	500	1.7e-2	1.7e-3	0.3
XGB _{6:PC} ^{AM}	dart	15	6.8e-6	5.7	0.8	1	400	6.3e-2	8.1e-2	6.9e-2
XGB ₆ ⁹⁴⁰	gbtree	20	1.0e-5	3	0.5	1	900	1.0e-2	7.2e-4	1.3e-3
XGB _{6:PC} ⁹⁴⁰	dart	15	4.3e-6	9.9	0.7	1	500	4.7e-2	4.9e-4	3.3e-2
XGB ₇ ^{AM}	dart	15	1.4e-5	3.9	0.3	1	700	5.6e-2	2.0e-3	5.9
XGB _{7:PC} ^{AM}	gbtree	20	4.7e-6	12.6	0.6	0.9	900	4.4e-2	7.9e-2	0.3
XGB ₇ ⁹⁴⁰	dart	20	2.5e-6	8.3	0.5	1	900	8.4e-2	1.1e-3	7.4
XGB _{7:PC} ⁹⁴⁰	gbtree	15	1.3e-4	1.5	0.4	1	600	2.4e-2	1.2e-4	4.9e-2

Table S5.9. Chosen hyperparameter combinations for SVR models

Model	Hyperparameter				
	kernel	C0	Tolerance	C	Epsilon
SVR ₄ ^{AM}	rbf	0.78	3.3e-05	15	0.15
SVR _{4:PC} ^{AM}	rbf	0.81	1.1e-03	15	0.15
SVR ₄ ⁹⁴⁰	rbf	0.31	2.3e-03	15	0.11
SVR _{4:PC} ⁹⁴⁰	rbf	0.81	1.4e-03	15	0.09
SVR ₅ ^{AM}	rbf	0.56	2.7e-04	15	0.16
SVR _{5:PC} ^{AM}	rbf	0.88	4.5e-03	15	0.19
SVR ₅ ⁹⁴⁰	rbf	0.67	1.8e-04	15	0.15
SVR _{5:PC} ⁹⁴⁰	rbf	0.69	3.8e-04	15	0.17
SVR ₆ ^{AM}	rbf	0.35	6.3e-03	15	0.19
SVR _{6:PC} ^{AM}	rbf	0.11	1.9e-04	15	0.24
SVR ₆ ⁹⁴⁰	rbf	0.94	2.9e-04	15	0.16
SVR _{6:PC} ⁹⁴⁰	rbf	0.42	1.7e-04	15	0.19
SVR ₇ ^{AM}	rbf	0.42	4.8e-04	15	0.22
SVR _{7:PC} ^{AM}	rbf	0.02	9.5e-04	15	0.19
SVR ₇ ⁹⁴⁰	rbf	0.82	7.2e-04	15	0.19
SVR _{7:PC} ⁹⁴⁰	rbf	0.95	1.2e-04	15	0.17

Table S5.10. Chosen hyperparameter combinations for DNN models

Мережа	Параметр								
	config	N_{hl}	N_{nll}	BS	Epochs	LR, 10^{-4}	Optim	Activ	Init
DNN ₄ ^{AM}	Pipe	8	50	64	500	12.0	Adamax	tanh	XN
DNN _{4:PC} ^{AM}	Pipe	8	120	16	500	5.4	Nadam	tanh	XU
DNN ₄ ⁹⁴⁰	Pipe	5	200	128	500	3.4	Nadam	tanh	XU
DNN _{4:PC} ⁹⁴⁰	Trapezium	-	250	16	500	3.7	Adamax	tanh	XN
DNN ₅ ^{AM}	Trapezium	-	50	16	500	2.7	Adam	tanh	XN
DNN _{5:PC} ^{AM}	Pipe	10	50	16	500	6.3	Adamax	relu	XN
DNN ₅ ⁹⁴⁰	Trapezium	-	200	64	500	7.8	Adamax	tanh	XU
DNN _{5:PC} ⁹⁴⁰	Pipe	5	100	16	500	1.9	Adamax	relu	XN
DNN ₆ ^{AM}	Trapezium	-	150	8	500	3.5	Nadam	tanh	XN
DNN _{6:PC} ^{AM}	Trapezium	-	150	128	500	1.5	Nadam	tanh	XU
DNN ₆ ⁹⁴⁰	Pipe	5	50	128	500	3.9	Nadam	elu	XU
DNN _{6:PC} ⁹⁴⁰	Trapezium	-	150	64	500	1.0	Nadam	tanh	XU
DNN ₇ ^{AM}	Pipe	5	120	16	500	10.3	Adamax	elu	XN
DNN _{7:PC} ^{AM}	Trapezium	-	100	32	500	18.3	Adamax	relu	XN
DNN ₇ ⁹⁴⁰	Pipe	6	50	16	500	16.4	Nadam	tanh	XN
DNN _{7:PC} ⁹⁴⁰	Pipe	6	100	8	500	60.5	Adamax	elu	XU

Table S5.11. Performance metrics of the models using fivefold cross-validation of train dataset

Model	MSE, 10^{-3}	MAPE, %	R^2 , 10^{-3}	Model	MSE, 10^{-3}	MAPE, %	R^2 , 10^{-3}
RF ₄ ⁹⁴⁰	5.2 ± 0.1	11 ± 2	980 ± 5	RF ₄ ^{AM}	32 ± 3	41 ± 7	970 ± 10
RF _{4:PC} ⁹⁴⁰	7.0 ± 1.0	12 ± 1	976 ± 4	RF _{4:PC} ^{AM}	52 ± 5	90 ± 30	910 ± 10
RF ₅ ⁹⁴⁰	4.1 ± 0.5	9.6 ± 0.5	994 ± 1	RF ₅ ^{AM}	12 ± 2	17 ± 3	993 ± 3
RF _{5:PC} ⁹⁴⁰	6 ± 1	13 ± 1	970 ± 10	RF _{5:PC} ^{AM}	76 ± 6	100 ± 20	895 ± 15
RF ₆ ⁹⁴⁰	3.0 ± 0.8	7.1 ± 0.5	997 ± 2	RF ₆ ^{AM}	4.3 ± 0.4	7.9 ± 0.5	994 ± 4
RF _{6:PC} ⁹⁴⁰	6 ± 1	14 ± 4	971 ± 8	RF _{6:PC} ^{AM}	25 ± 5	34 ± 10	940 ± 15
RF ₇ ⁹⁴⁰	3.0 ± 0.8	6.6 ± 0.3	998 ± 1	RF ₇ ^{AM}	3.9 ± 0.4	7.0 ± 0.7	994 ± 6
RF _{7:PC} ⁹⁴⁰	3.6 ± 0.9	9.0 ± 0.5	993 ± 2	RF _{7:PC} ^{AM}	7 ± 1	12 ± 1	966 ± 5
GB ₄ ⁹⁴⁰	3.7 ± 0.9	7.4 ± 0.7	986 ± 3	GB ₄ ^{AM}	34 ± 6	60 ± 20	940 ± 7
GB _{4:PC} ⁹⁴⁰	5.0 ± 0.9	10 ± 1	975 ± 8	GB _{4:PC} ^{AM}	55 ± 8	100 ± 30	910 ± 10
GB ₅ ⁹⁴⁰	2.5 ± 0.5	6.5 ± 0.2	995 ± 1	GB ₅ ^{AM}	10 ± 2	14 ± 4	993 ± 3
GB _{5:PC} ⁹⁴⁰	4 ± 1	9.4 ± 0.7	983 ± 4	GB _{5:PC} ^{AM}	77 ± 7	130 ± 10	900 ± 15
GB ₆ ⁹⁴⁰	1.9 ± 0.3	5.2 ± 0.3	997 ± 1	GB ₆ ^{AM}	4.1 ± 0.6	7.7 ± 0.9	997 ± 2
GB _{6:PC} ⁹⁴⁰	5 ± 1	11 ± 4	976 ± 7	GB _{6:PC} ^{AM}	23 ± 4	30 ± 10	937 ± 7
GB ₇ ⁹⁴⁰	1.9 ± 0.2	5.3 ± 0.3	998 ± 1	GB ₇ ^{AM}	3.3 ± 0.6	6.5 ± 0.4	992 ± 4
GB _{7:PC} ⁹⁴⁰	3.2 ± 0.6	8.0 ± 0.6	992 ± 3	GB _{7:PC} ^{AM}	5.4 ± 0.9	9.6 ± 0.5	970 ± 10
XGB ₄ ⁹⁴⁰	4.8 ± 0.5	9.2 ± 0.8	964 ± 4	XGB ₄ ^{AM}	36 ± 4	50 ± 10	925 ± 15
XGB _{4:PC} ⁹⁴⁰	8.7 ± 0.5	15 ± 1	960 ± 6	XGB _{4:PC} ^{AM}	52 ± 5	110 ± 50	900 ± 10
XGB ₅ ⁹⁴⁰	2.8 ± 0.4	6.4 ± 0.4	982 ± 4	XGB ₅ ^{AM}	10 ± 2	19 ± 7	985 ± 2
XGB _{5:PC} ⁹⁴⁰	6.5 ± 0.4	11.2 ± 0.3	966 ± 6	XGB _{5:PC} ^{AM}	80 ± 4	130 ± 30	870 ± 10
XGB ₆ ⁹⁴⁰	1.4 ± 0.3	4.3 ± 0.3	996 ± 1	XGB ₆ ^{AM}	3.3 ± 0.5	6.8 ± 0.7	997 ± 3
XGB _{6:PC} ⁹⁴⁰	5.8 ± 0.9	11 ± 1	968 ± 6	XGB _{6:PC} ^{AM}	22 ± 2	35 ± 10	950 ± 10
XGB ₇ ⁹⁴⁰	1.5 ± 0.2	5.4 ± 0.1	996 ± 1	XGB ₇ ^{AM}	2.7 ± 0.3	6.5 ± 0.6	992 ± 4
XGB _{7:PC} ⁹⁴⁰	4 ± 1	8.1 ± 0.5	970 ± 20	XGB _{7:PC} ^{AM}	5 ± 1	10 ± 1	961 ± 7
SVR ₄ ⁹⁴⁰	220 ± 20	200 ± 20	540 ± 30	SVR ₄ ^{AM}	230 ± 5	220 ± 20	500 ± 20
SVR _{4:PC} ⁹⁴⁰	221 ± 7	205 ± 15	530 ± 40	SVR _{4:PC} ^{AM}	230 ± 10	215 ± 10	510 ± 20
SVR ₅ ⁹⁴⁰	210 ± 10	180 ± 15	480 ± 40	SVR ₅ ^{AM}	200 ± 10	180 ± 30	520 ± 20
SVR _{5:PC} ⁹⁴⁰	215 ± 9	180 ± 15	520 ± 20	SVR _{5:PC} ^{AM}	243 ± 8	220 ± 8	500 ± 20
SVR ₆ ⁹⁴⁰	180 ± 7	127 ± 7	600 ± 10	SVR ₆ ^{AM}	180 ± 15	59 ± 2	420 ± 30
SVR _{6:PC} ⁹⁴⁰	204 ± 8	150 ± 10	515 ± 20	SVR _{6:PC} ^{AM}	210 ± 15	76 ± 4	300 ± 60
SVR ₇ ⁹⁴⁰	161 ± 9	115 ± 10	610 ± 30	SVR ₇ ^{AM}	140 ± 8	55 ± 1	390 ± 60

SVR ⁹⁴⁰ _{7:PC}	188 ± 9	124 ± 8	600 ± 20	SVR ^{AM} _{7:PC}	140 ± 9	50 ± 1	450 ± 10
DNN ⁹⁴⁰ ₄	6 ± 1	10 ± 2	971 ± 3	DNN ^{AM} ₄	38 ± 4	57 ± 3	940 ± 10
DNN ⁹⁴⁰ _{4:PC}	6 ± 2	9 ± 2	980 ± 10	DNN ^{AM} _{4:PC}	80 ± 20	125 ± 40	860 ± 30
DNN ⁹⁴⁰ ₅	6 ± 1	11 ± 2	970 ± 7	DNN ^{AM} ₅	9 ± 6	25 ± 15	985 ± 5
DNN ⁹⁴⁰ _{5:PC}	7 ± 2	12 ± 3	973 ± 6	DNN ^{AM} _{5:PC}	80 ± 8	230 ± 110	930 ± 15
DNN ⁹⁴⁰ ₆	4 ± 1	11 ± 3	970 ± 9	DNN ^{AM} ₆	10 ± 6	14 ± 5	980 ± 15
DNN ⁹⁴⁰ _{6:PC}	5 ± 2	12 ± 7	967 ± 6	DNN ^{AM} _{6:PC}	2.2 ± 0.7	7 ± 2	994 ± 4
DNN ⁹⁴⁰ ₇	20 ± 10	45 ± 25	880 ± 100	DNN ^{AM} ₇	0.8 ± 0.4	3.2 ± 0.7	997 ± 2
DNN ⁹⁴⁰ _{7:PC}	15 ± 10	14 ± 2	975 ± 7	DNN ^{AM} _{7:PC}	0.9 ± 0.5	3 ± 1	997 ± 2

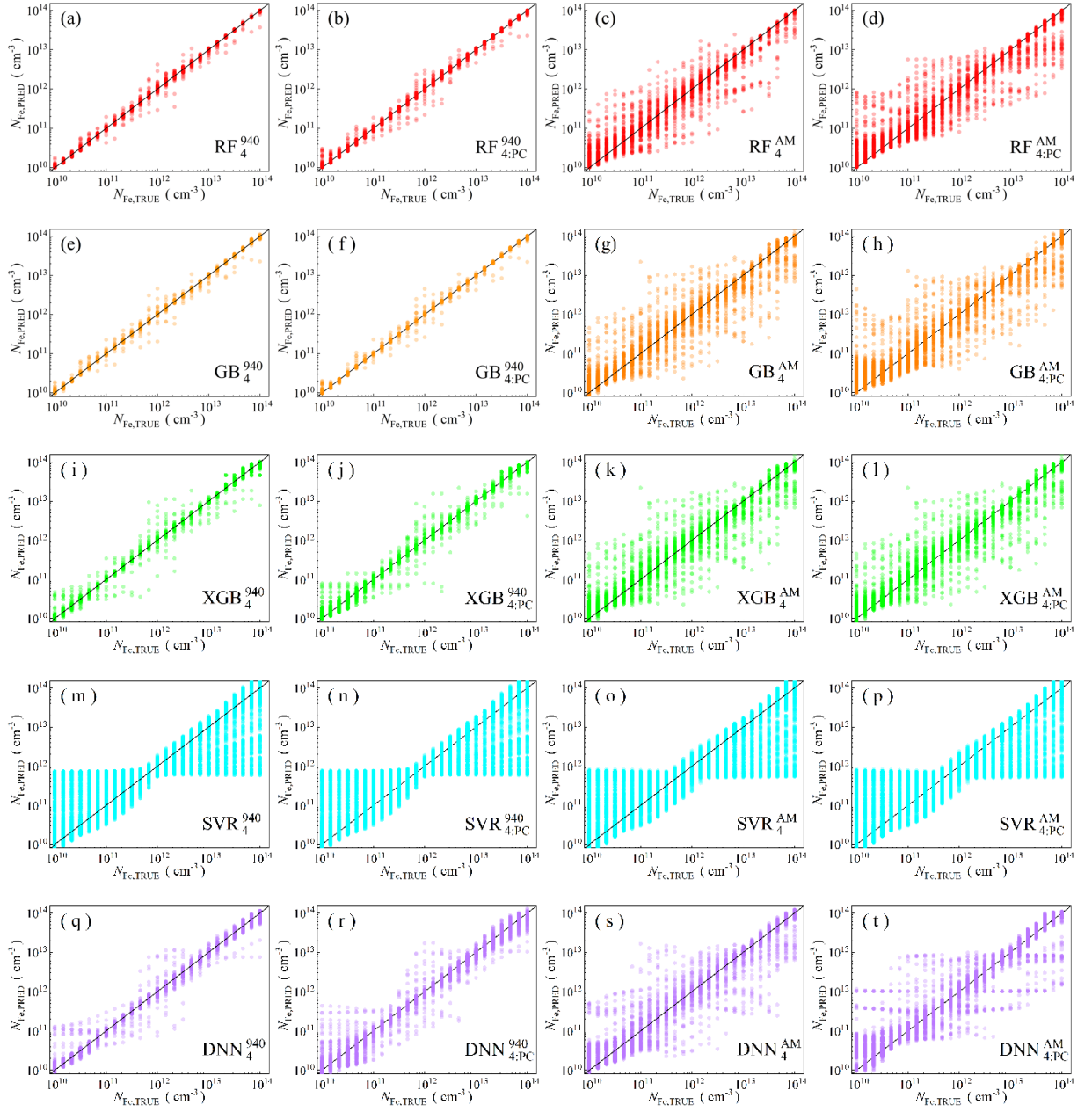


Fig.S5.3. Scatter plots compare reference iron concentrations with ML-predicted values during the training phase in the case of 4D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

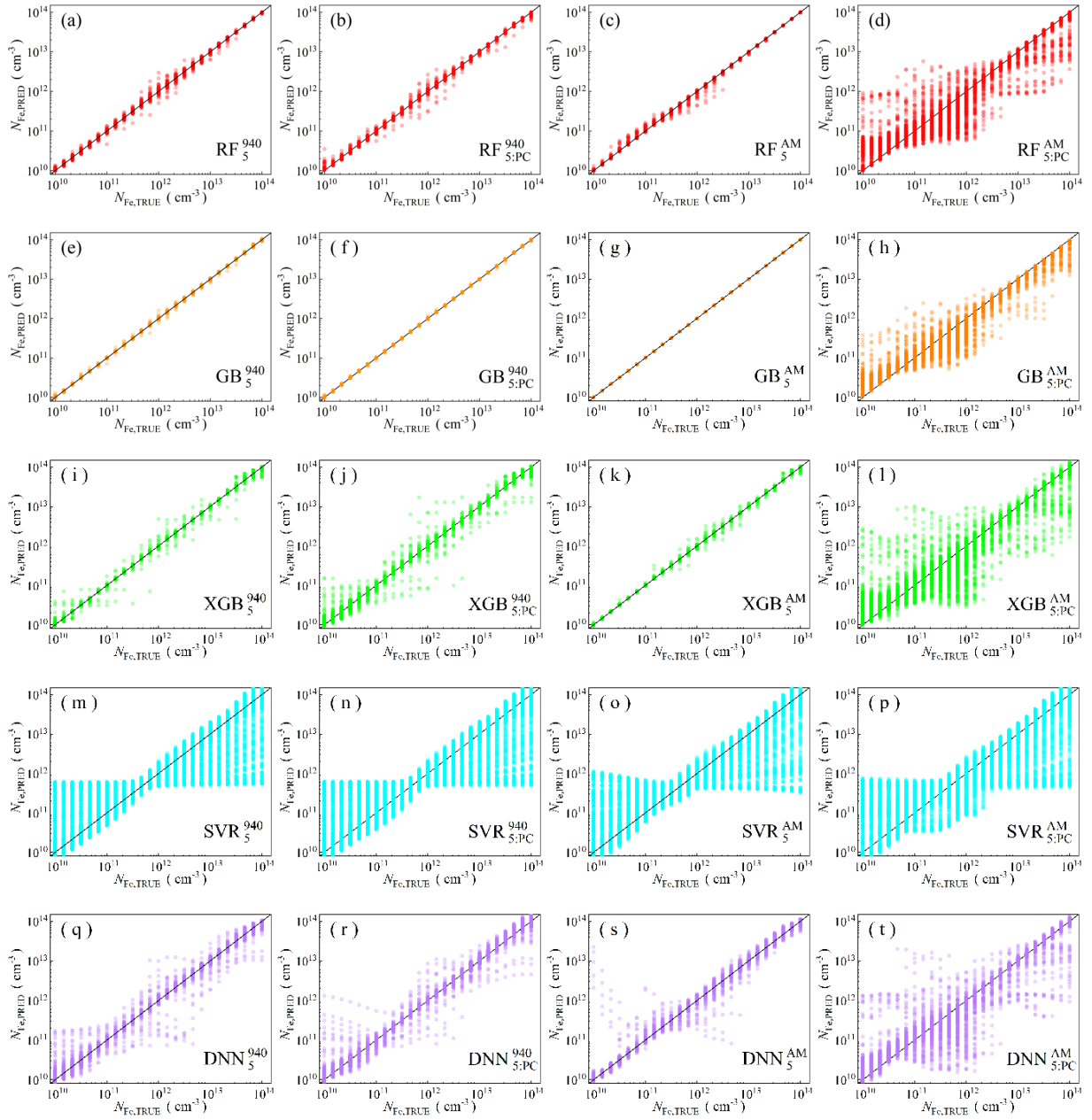


Fig.S5.4. Scatter plots compare reference iron concentrations with ML-predicted values during the training phase in the case of 5D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

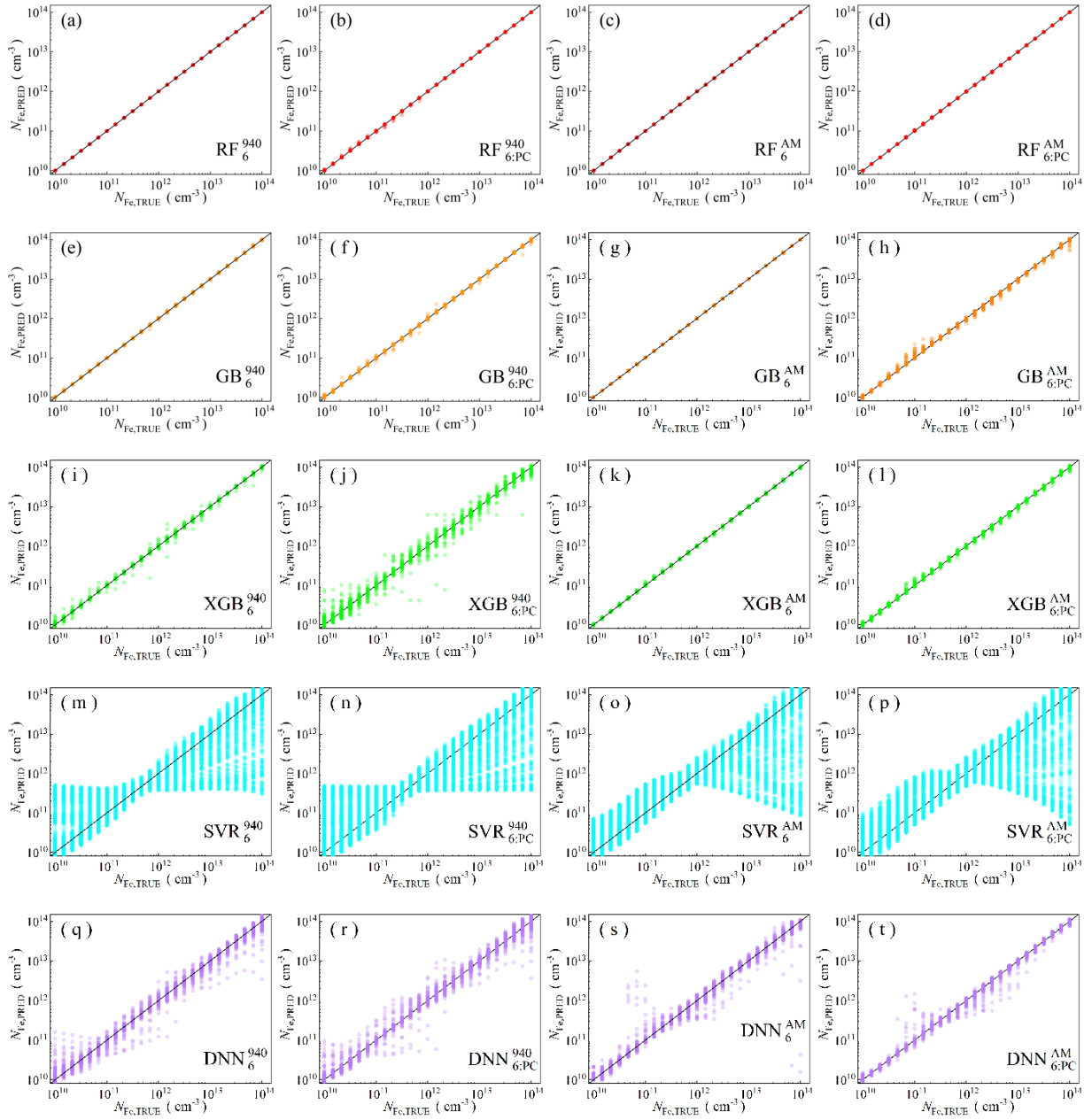


Fig.S5.5. Scatter plots compare reference iron concentrations with ML-predicted values during the training phase in the case of 6D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

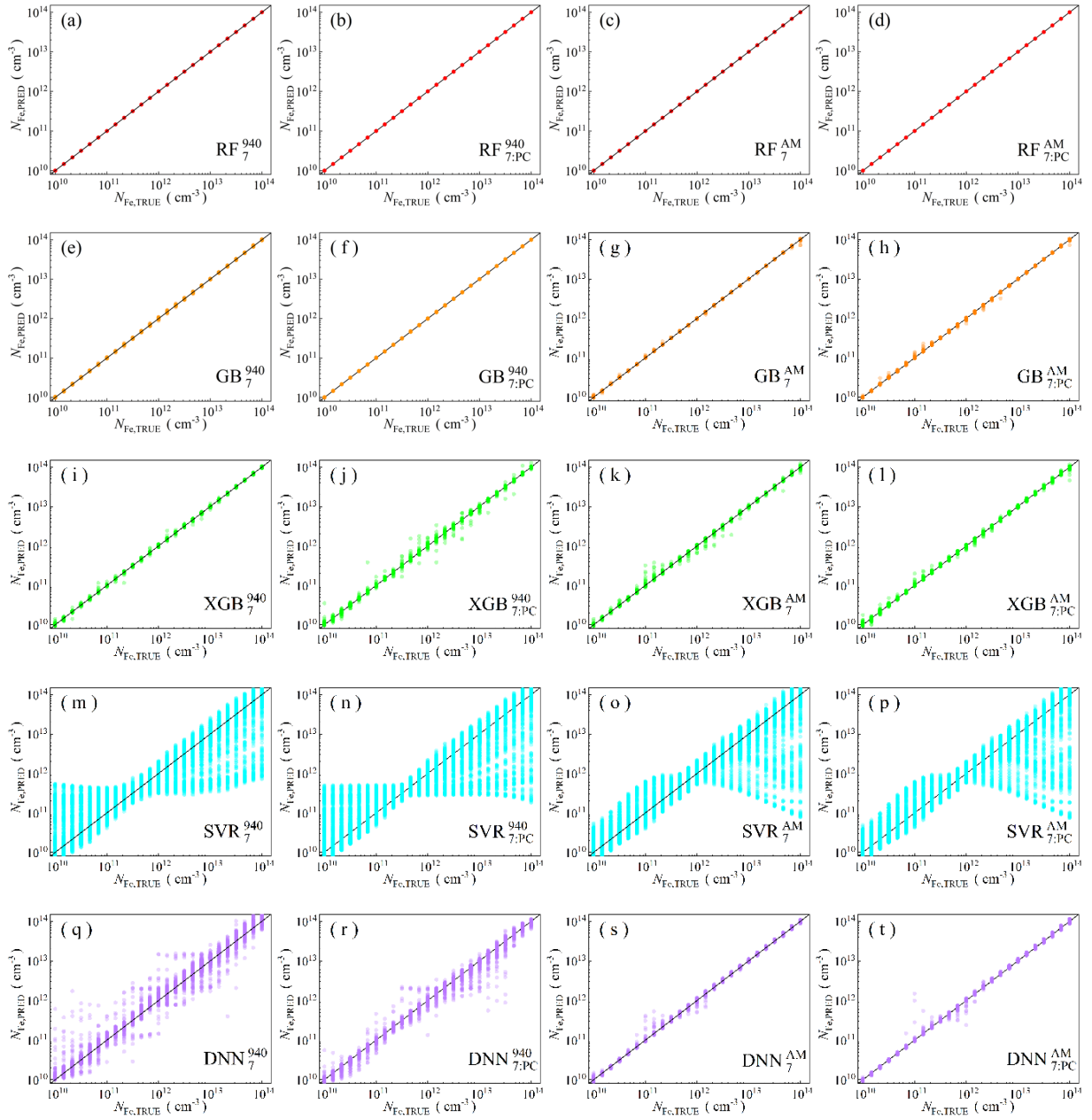


Fig.S5.6. Scatter plots compare reference iron concentrations with ML-predicted values during the training phase in the case of 7D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

Table S5.12. Performance metrics of the models for train dataset. Illumination 940 nm.

Model	MSE, 10^{-3}	R ²	MAPE, %	MdAPE, %	p01, %	p10, %
RF ₄ ⁹⁴⁰	0.60	0.9979	3.11	1.468	39.85	95.37
RF _{4:PC} ⁹⁴⁰	1.05	0.99639	4.20	2.134	32.19	91.22
RF ₅ ⁹⁴⁰	0.444	0.99976	2.91	1.077	48.39	94.63
RF _{5:PC} ⁹⁴⁰	0.662	0.9967	3.89	2.174	29.56	92.1
RF ₆ ⁹⁴⁰	0	1.000	0.002	8.2E-13	100	100
RF _{6:PC} ⁹⁴⁰	0.0036	0.999	0.045	8.2E-13	99.47	99.91
RF ₇ ⁹⁴⁰	0	1.000	0	8.2E-13	100	100
RF _{7:PC} ⁹⁴⁰	0	1.000	0.0026	8.2E-13	99.96	100
GB ₄ ⁹⁴⁰	0.499	0.99725	0.160	0.5048	71.02	97.87
GB _{4:PC} ⁹⁴⁰	0.47	0.9977	1.52	0.4638	72.64	97.98
GB ₅ ⁹⁴⁰	0.0402	0.99970	0.604	0.2315	87.26	99.47
GB _{5:PC} ⁹⁴⁰	0.0087	0.99993	0.36	0.1678	93.65	99.93
GB ₆ ⁹⁴⁰	0.0058	0.999	0.26	0.1112	95.68	99.98
GB _{6:PC} ⁹⁴⁰	0.0370	0.99970	0.599	0.2469	86.68	99.56
GB ₇ ⁹⁴⁰	0.0119	0.99992	0.50	0.2419	87.62	99.9
GB _{7:PC} ⁹⁴⁰	0.0007	1.000	0.107	0.05002	99.14	100
XGB ₄ ⁹⁴⁰	2.282	0.98091	4.64	1.559	36	92.15
XGB _{4:PC} ⁹⁴⁰	3.929	0.9846	7.41	2.17	30.5	82.72
XGB ₅ ⁹⁴⁰	0.788	0.99490	1.94	0.617	66.3	97.31
XGB _{5:PC} ⁹⁴⁰	3.729	0.9857	6.56	1.85	32.43	88.35
XGB ₆ ⁹⁴⁰	0.174	0.99954	1.060	0.4432	76.06	98.93
XGB _{6:PC} ⁹⁴⁰	1.97	0.9924	4.63	1.645	34.76	91.14
XGB ₇ ⁹⁴⁰	0.052	0.99983	0.84	0.4336	77.72	99.52
XGB _{7:PC} ⁹⁴⁰	0.35	0.9992	2.01	1.156	44.52	98.1
SVR ₄ ⁹⁴⁰	215	0.549	201	33.24	1.358	15.48
SVR _{4:PC} ⁹⁴⁰	215	0.537	200	29.55	1.673	17.46
SVR ₅ ⁹⁴⁰	204	0.492	171	34.94	1.552	15.05
SVR _{5:PC} ⁹⁴⁰	210	0.535	177	40.48	1.139	12.77
SVR ₆ ⁹⁴⁰	174	0.609	124	35.86	1.608	13.71
SVR _{6:PC} ⁹⁴⁰	199	0.548	142	41.63	1.277	11.81
SVR ₇ ⁹⁴⁰	155	0.626	112	37.34	1.188	12.42
SVR _{7:PC} ⁹⁴⁰	182	0.628	122	36.34	1.349	14.22

DNN ⁹⁴⁰ ₄	5	0.962	12	6.787	7.717	67.28
DNN ⁹⁴⁰ _{4:PC}	11	0.952	17	7.833	6.804	60.34
DNN ⁹⁴⁰ ₅	8	0.963	12	5.977	8.727	72.91
DNN ⁹⁴⁰ _{5:PC}	8	0.940	17	6.983	7.572	65.73
DNN ⁹⁴⁰ ₆	4	0.966	9	5.205	10.26	75.68
DNN ⁹⁴⁰ _{6:PC}	4	0.958	9	4.611	12.86	77.29
DNN ⁹⁴⁰ ₇	12	0.931	24	14.4	4.186	38.54
DNN ⁹⁴⁰ _{7:PC}	4	0.983	8	5.109	10.03	77.92

Table S5.13. Performance metrics of the models for train dataset. Illumination AM1.5.

Model	MSE, 10 ⁻³	R ²	MAPE, %	MdAPE, %	p01, %	p10, %
RF ^{AM} ₄	15.7	0.986	16.3	2.264	37.96	78.81
RF ^{AM} _{4:PC}	27.6	0.9407	30.9	3.939	18.95	73.54
RF ^{AM} ₅	0.25	0.99993	1.0	0.1921	90.97	98.26
RF ^{AM} _{5:PC}	43.1	0.929	44.1	3.89	24.28	68.2
RF ^{AM} ₆	0.0009	0.99999	0.1	8.1E-13	100	100
RF ^{AM} _{6:PC}	0.0008	1.000	0.0032	1.6E-12	98.95	100
RF ^{AM} ₇	0	1.000	0	1.2E-12	100	100
RF ^{AM} _{7:PC}	0	1.000	0	8.2E-13	100	100
GB ^{AM} ₄	12.9	0.9693	16.3	4.521	10.55	81.41
GB ^{AM} _{4:PC}	19.4	0.9559	21.9	3.506	14.34	82.42
GB ^{AM} ₅	0	1.000	0.008	0.002666	100	100
GB ^{AM} _{5:PC}	20.3	0.9586	23.4	6.337	7.232	70.86
GB ^{AM} ₆	0.0002	1.000	0.0027	0.007811	99.77	100
GB ^{AM} _{6:PC}	0.00142	0.99858	1.01	0.3292	80.75	98.55
GB ^{AM} ₇	0.026	0.99984	0.48	0.1894	89.64	99.77
GB ^{AM} _{7:PC}	0.0078	0.99971	0.197	0.08304	97.52	99.88
XGB ^{AM} ₄	24.2	0.9465	26.9	4.362	14.63	73.26
XGB ^{AM} _{4:PC}	32.3	0.933	43.9	4.954	12.61	69.65
XGB ^{AM} ₅	1.820	0.99777	1.158	0.541	73.43	98.59
XGB ^{AM} _{5:PC}	55.0	0.906	66.4	7.317	9.891	57.49
XGB ^{AM} ₆	0.025	0.999	0.897	0.6342	68.15	99.85
XGB ^{AM} _{6:PC}	0.161	0.9983	2.27	1.592	33.94	98.52
XGB ^{AM} ₇	0.17	0.9991	1.38	0.731	61.61	98.93
XGB ^{AM} _{7:PC}	0.154	0.9979	2.04	1.389	38.34	98.63
SVR ^{AM} ₄	225	0.520	209	42.62	1.131	11.71
SVR ^{AM} _{4:PC}	224	0.521	209	41.86	1.366	12.08
SVR ^{AM} ₅	197	0.551	181	38.96	1.244	11.14
SVR ^{AM} _{5:PC}	238	0.525	213	50.62	0.897	9.939
SVR ^{AM} ₆	170	0.465	56	38.36	1.261	11.85
SVR ^{AM} _{6:PC}	202	0.336	74	46.59	0.9212	9.228
SVR ^{AM} ₇	133	0.444	53	38.09	1.172	12.66
SVR ^{AM} _{7:PC}	133	0.504	49	34.6	1.584	14.17

$\text{DNN}_{4}^{\text{AM}}$	31	0.936	37	10.17	5.067	49.32
$\text{DNN}_{4:\text{PC}}^{\text{AM}}$	47	0.870	61	14.32	3.79	36.55
$\text{DNN}_{5}^{\text{AM}}$	8	0.987	61	5.279	10.41	75.47
$\text{DNN}_{5:\text{PC}}^{\text{AM}}$	48	0.941	57	6.544	9.156	63.31
$\text{DNN}_{6}^{\text{AM}}$	6	0.979	11	5.287	10.69	75.45
$\text{DNN}_{6:\text{PC}}^{\text{AM}}$	2	0.995	5	3.01	19.61	92.74
$\text{DNN}_{7}^{\text{AM}}$	0.3	0.998	3	1.747	29.81	98
$\text{DNN}_{7:\text{PC}}^{\text{AM}}$	0.4	0.998	3	2.146	24.64	97.9

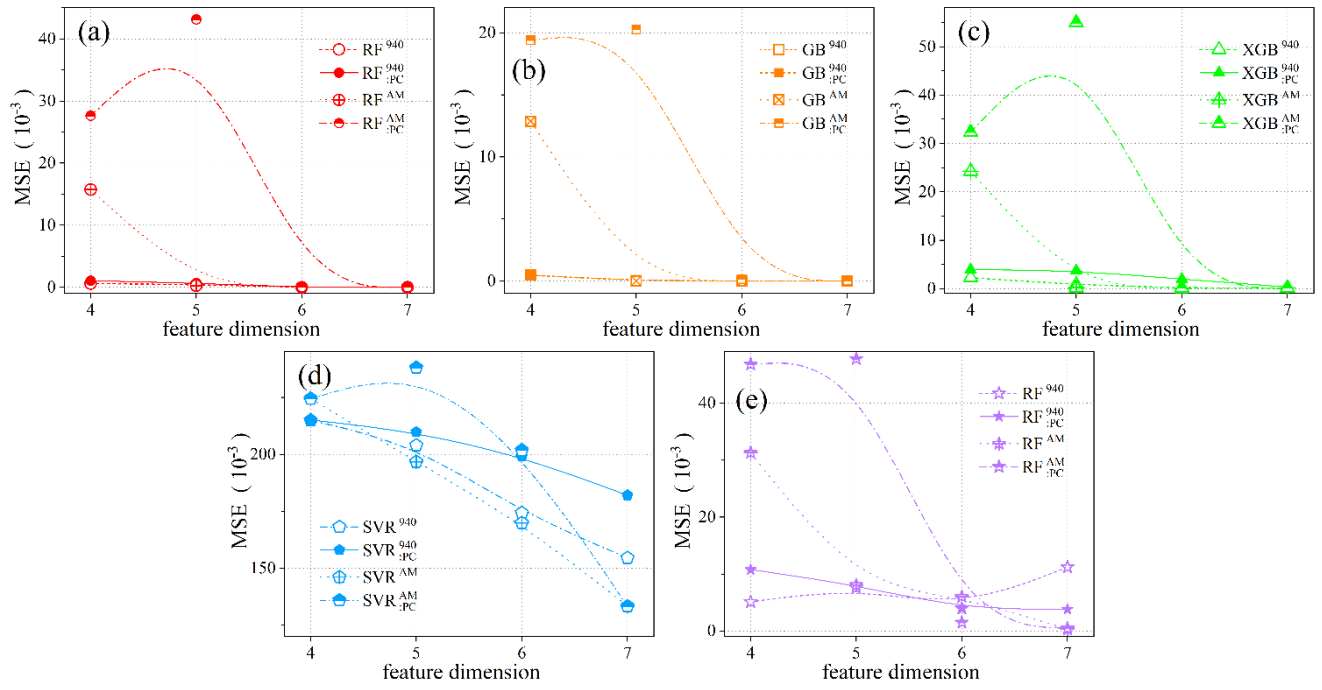


Fig.S5.7. Dependencies of MSE on the input feature dimension for train data.

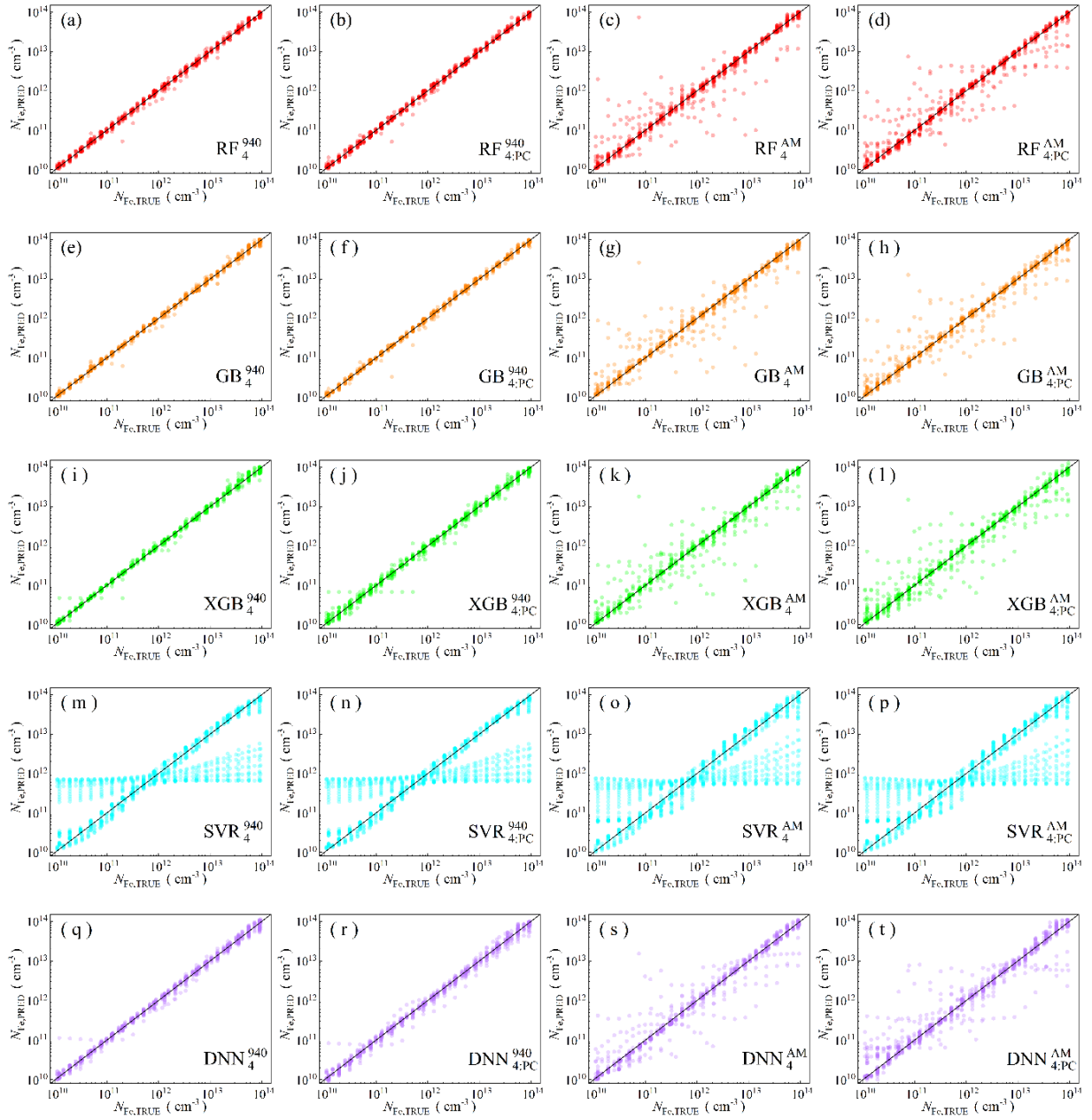


Fig.S5.8. Scatter plots compare reference iron concentrations with ML-predicted values during the N_{Fe} -altered test in the case of 4D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

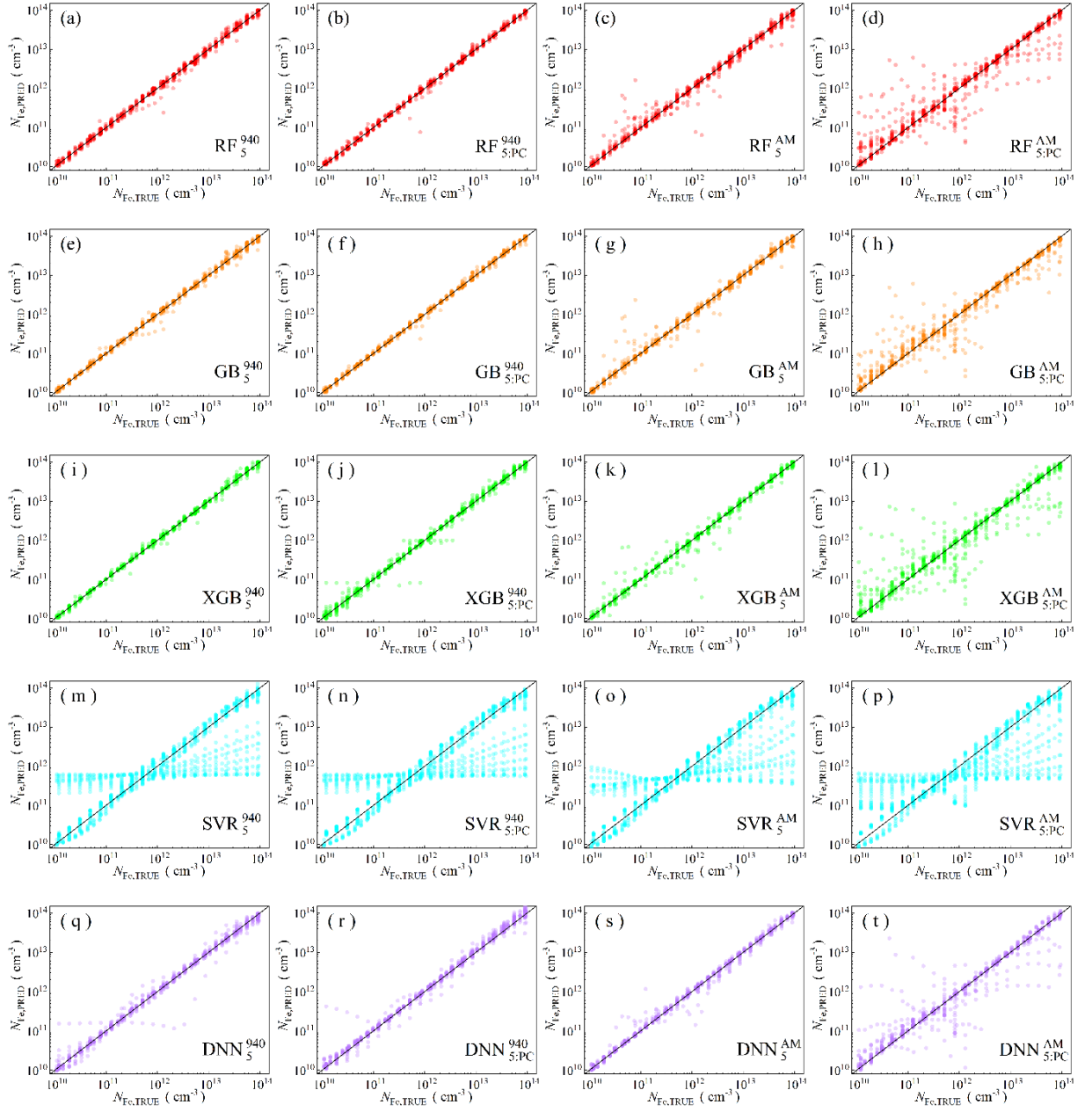


Fig.S5.9. Scatter plots compare reference iron concentrations with ML-predicted values during the N_{Fe} -altered test in the case of 5D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

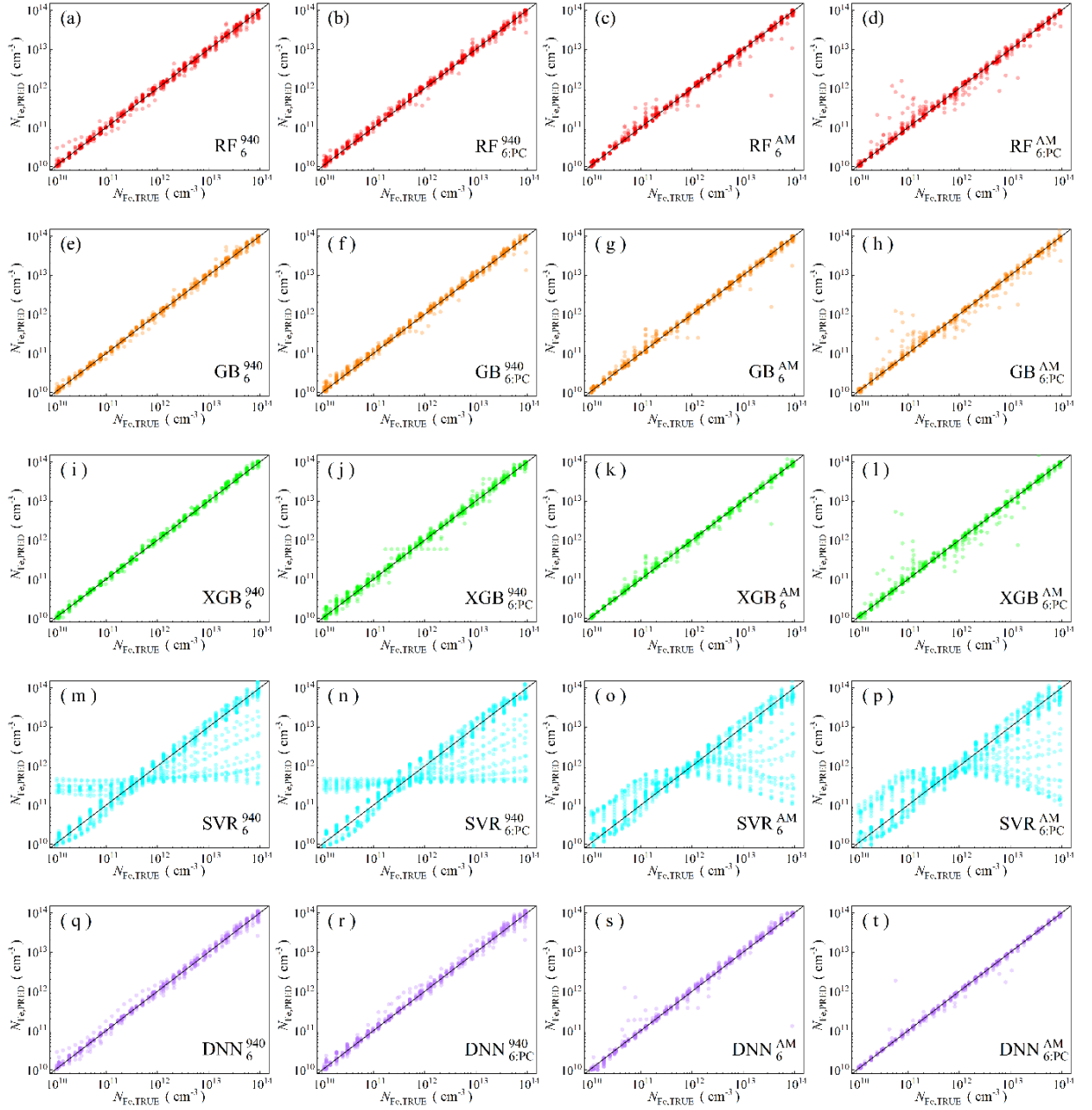


Fig.S5.10. Scatter plots compare reference iron concentrations with ML-predicted values during the N_{Fe} -altered test in the case of 6D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

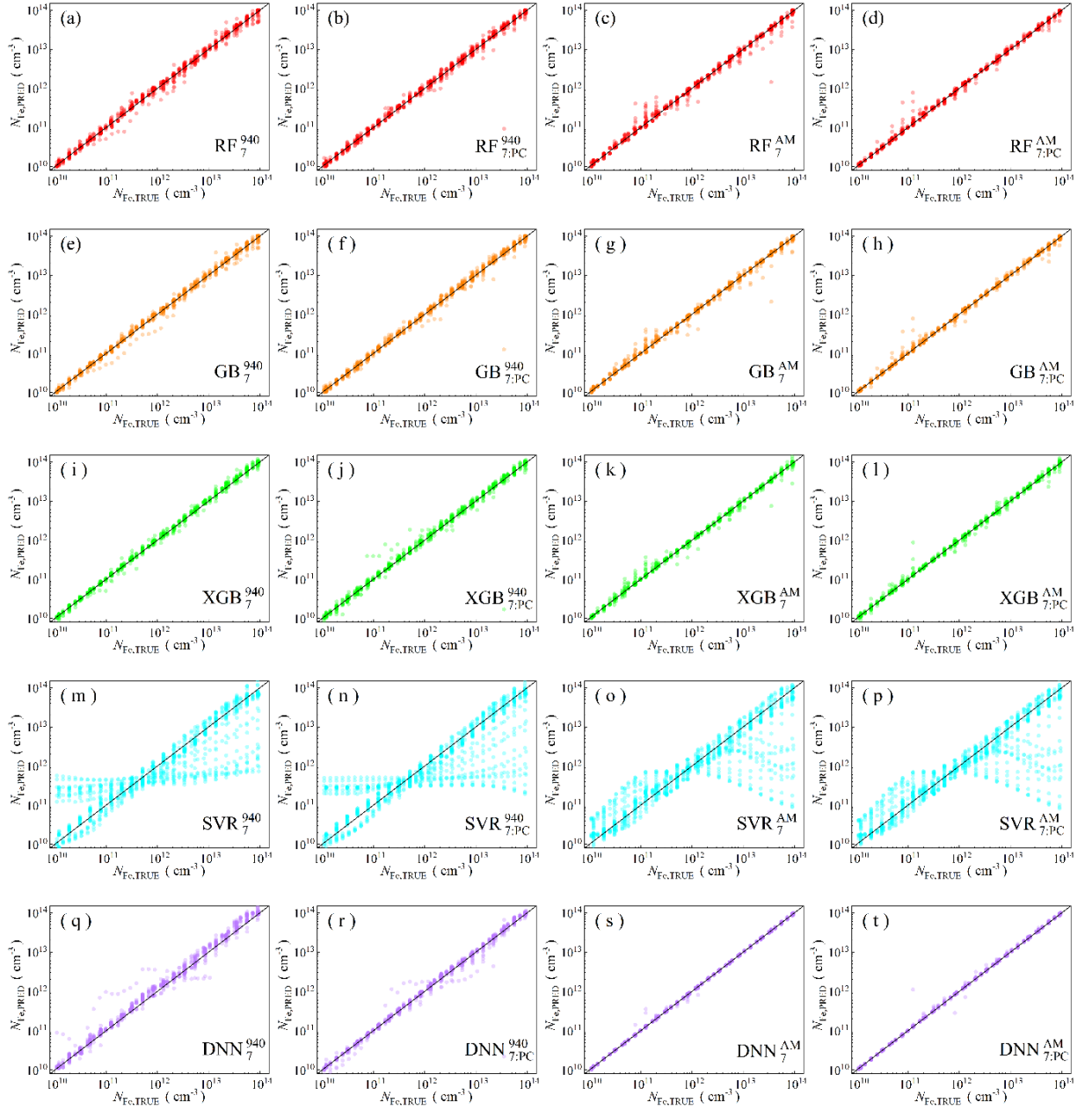


Fig.S5.11. Scatter plots compare reference iron concentrations with ML-predicted values during the N_{Fe} -altered test in the case of 7D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

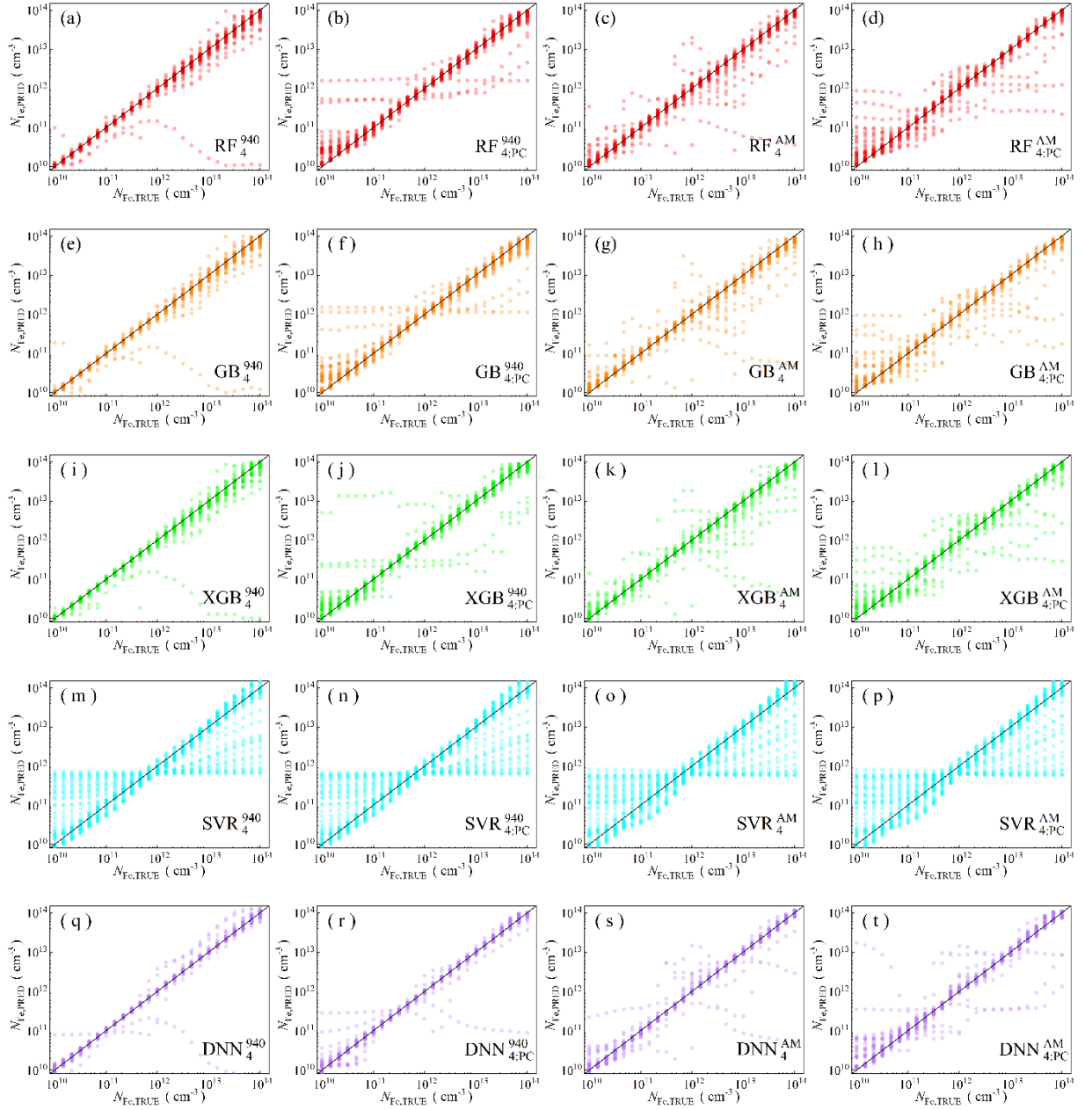


Fig.S5.12. Scatter plots compare reference iron concentrations with ML-predicted values during the T -altered test in the case of 4D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

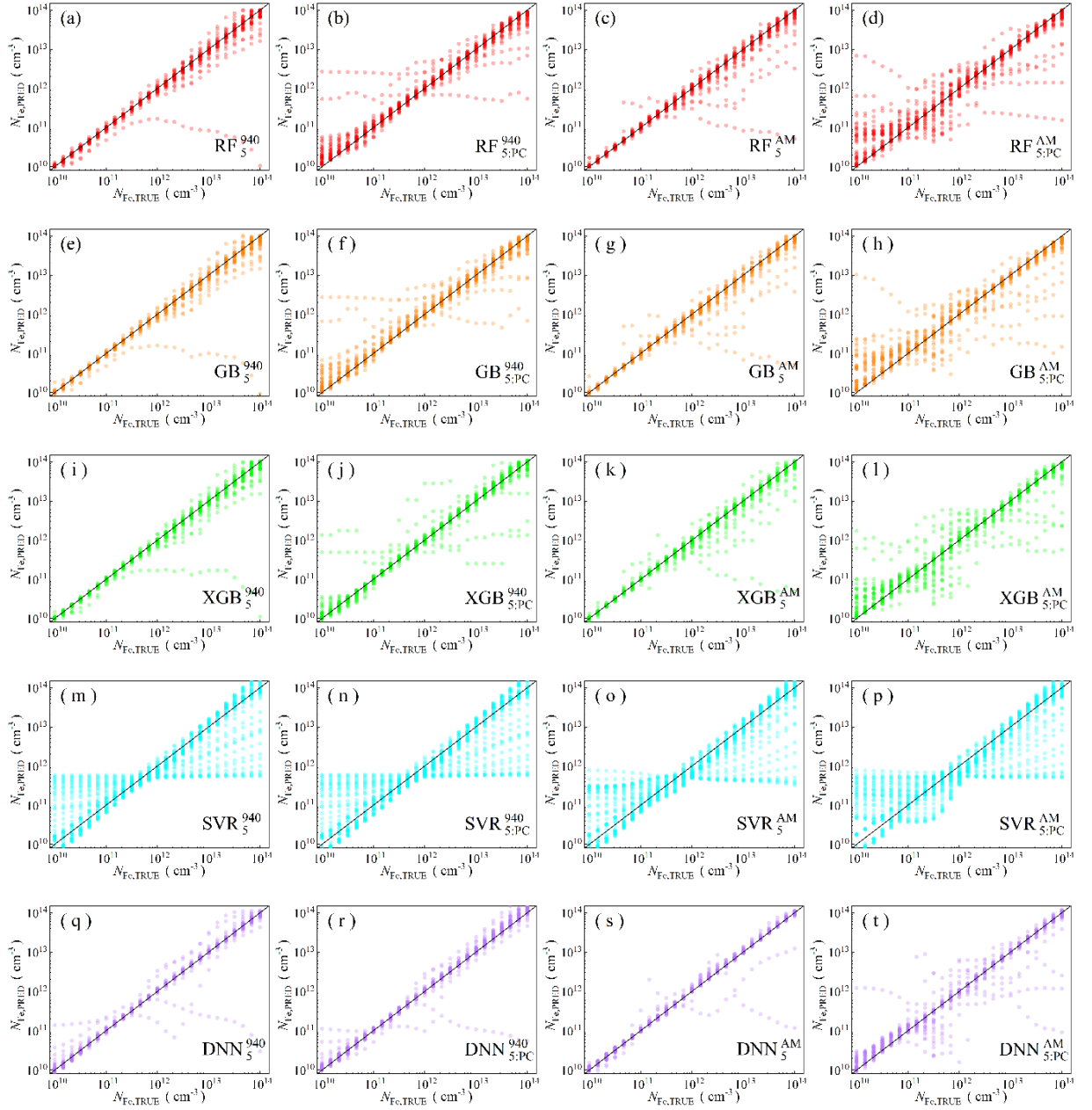


Fig.S5.13. Scatter plots compare reference iron concentrations with ML-predicted values during the T -altered test in the case of 5D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

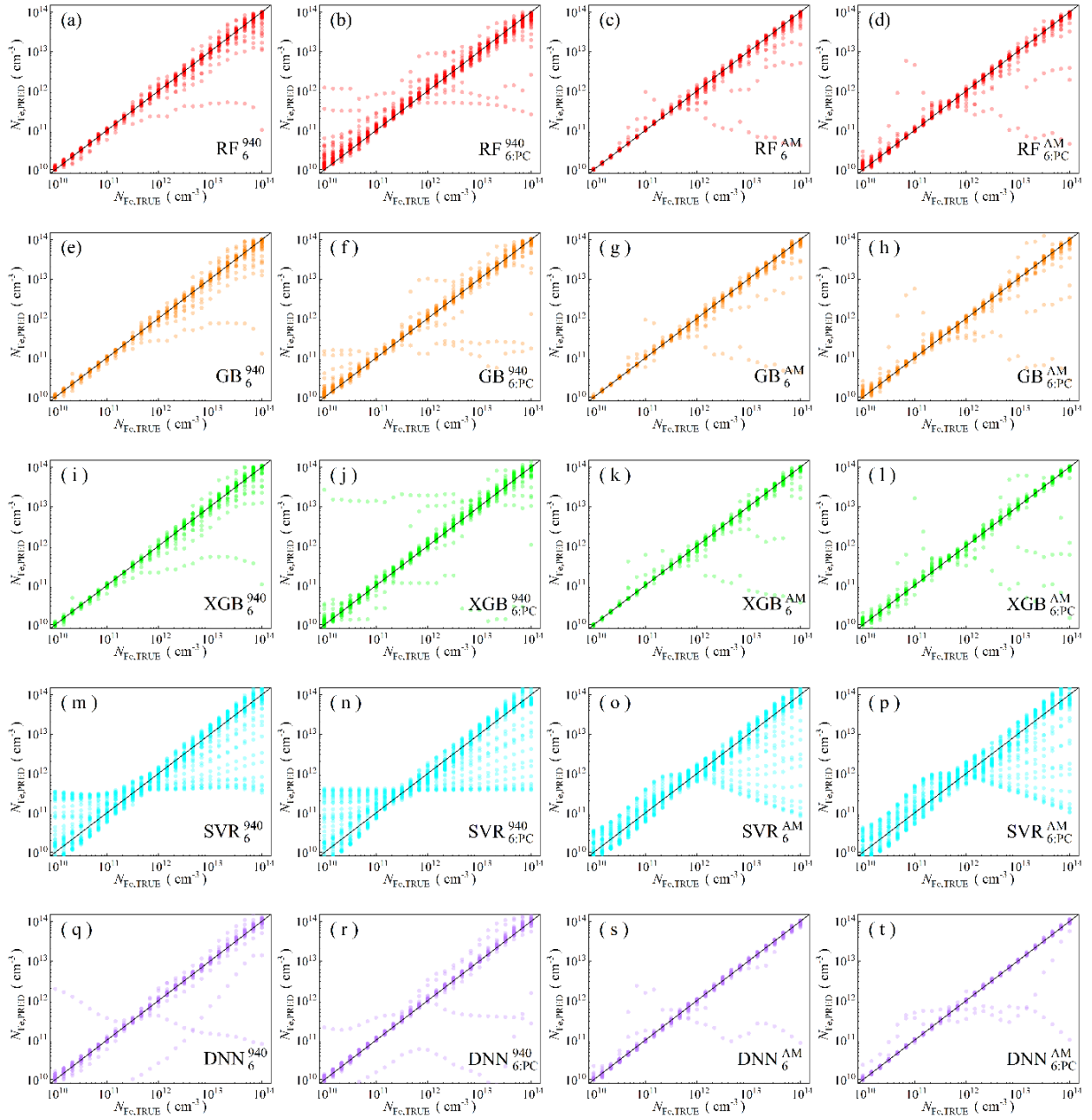


Fig.S5.14. Scatter plots compare reference iron concentrations with ML-predicted values during the T -altered test in the case of 6D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

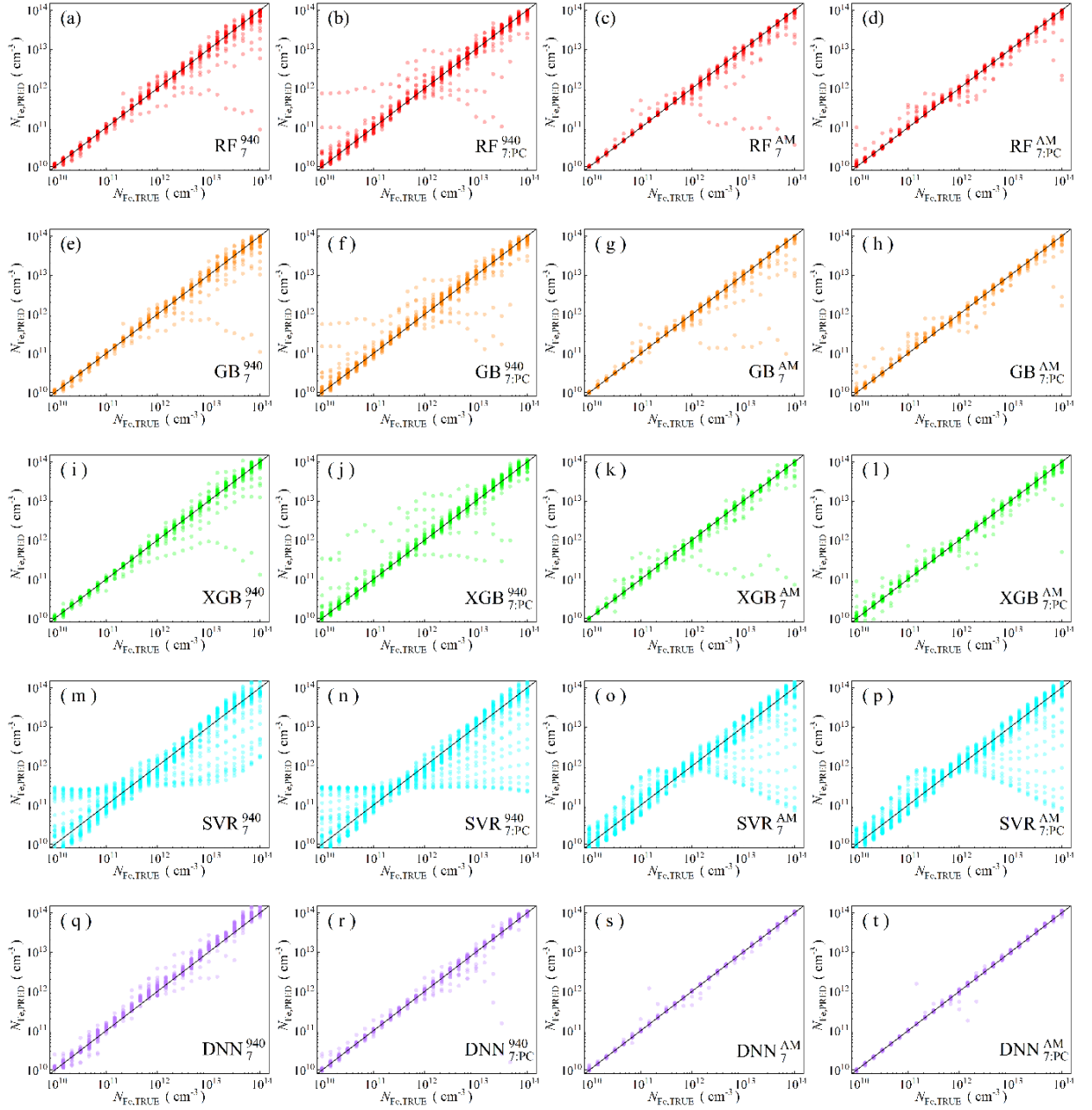


Fig.S5.15. Scatter plots compare reference iron concentrations with ML-predicted values during the T -altered test in the case of 7D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references..

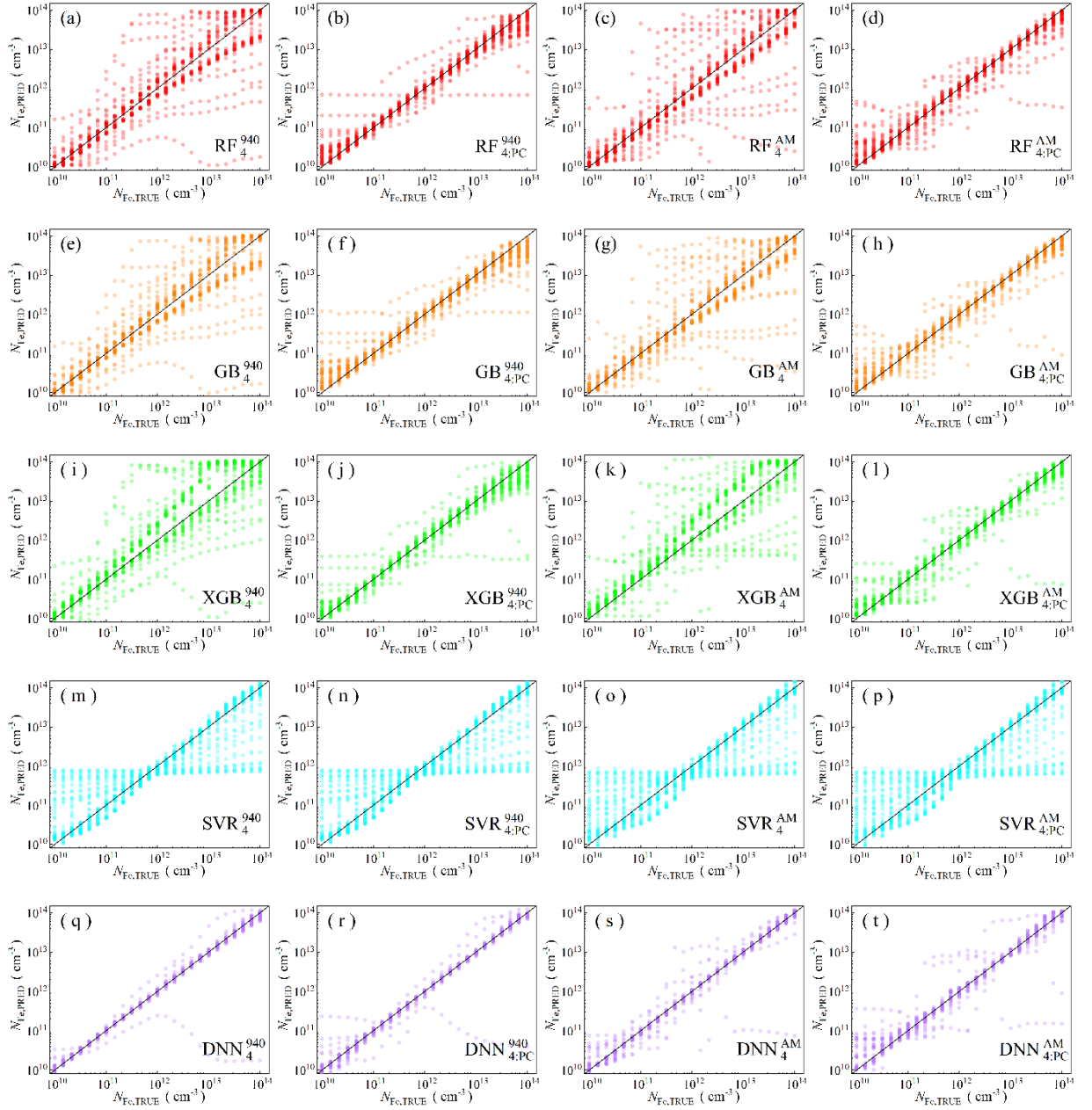


Fig.S5.16. Scatter plots compare reference iron concentrations with ML-predicted values during the N_B -altered test in the case of 4D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

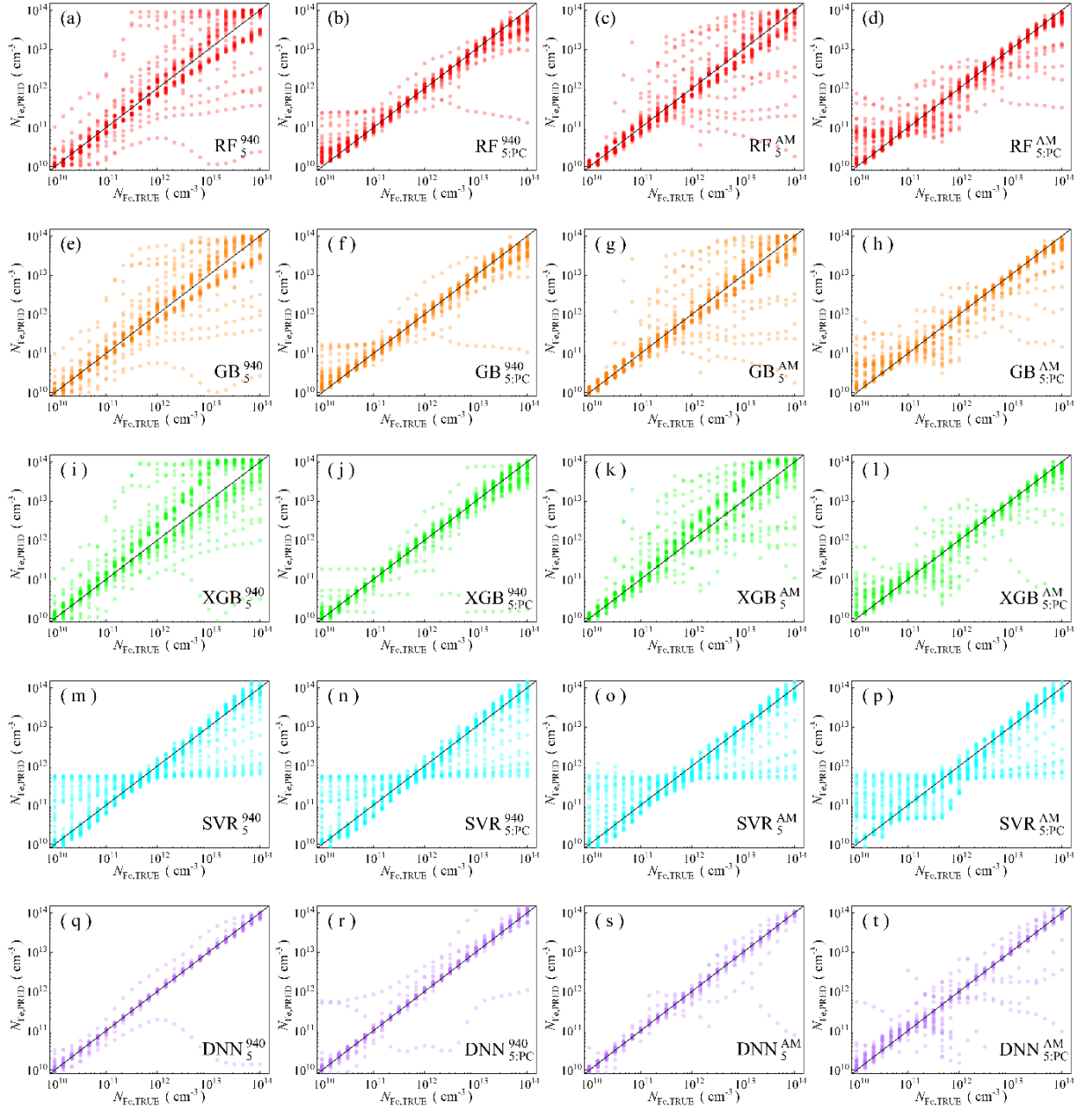


Fig.S5.17. Scatter plots compare reference iron concentrations with ML-predicted values during the N_B -altered test in the case of 5D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

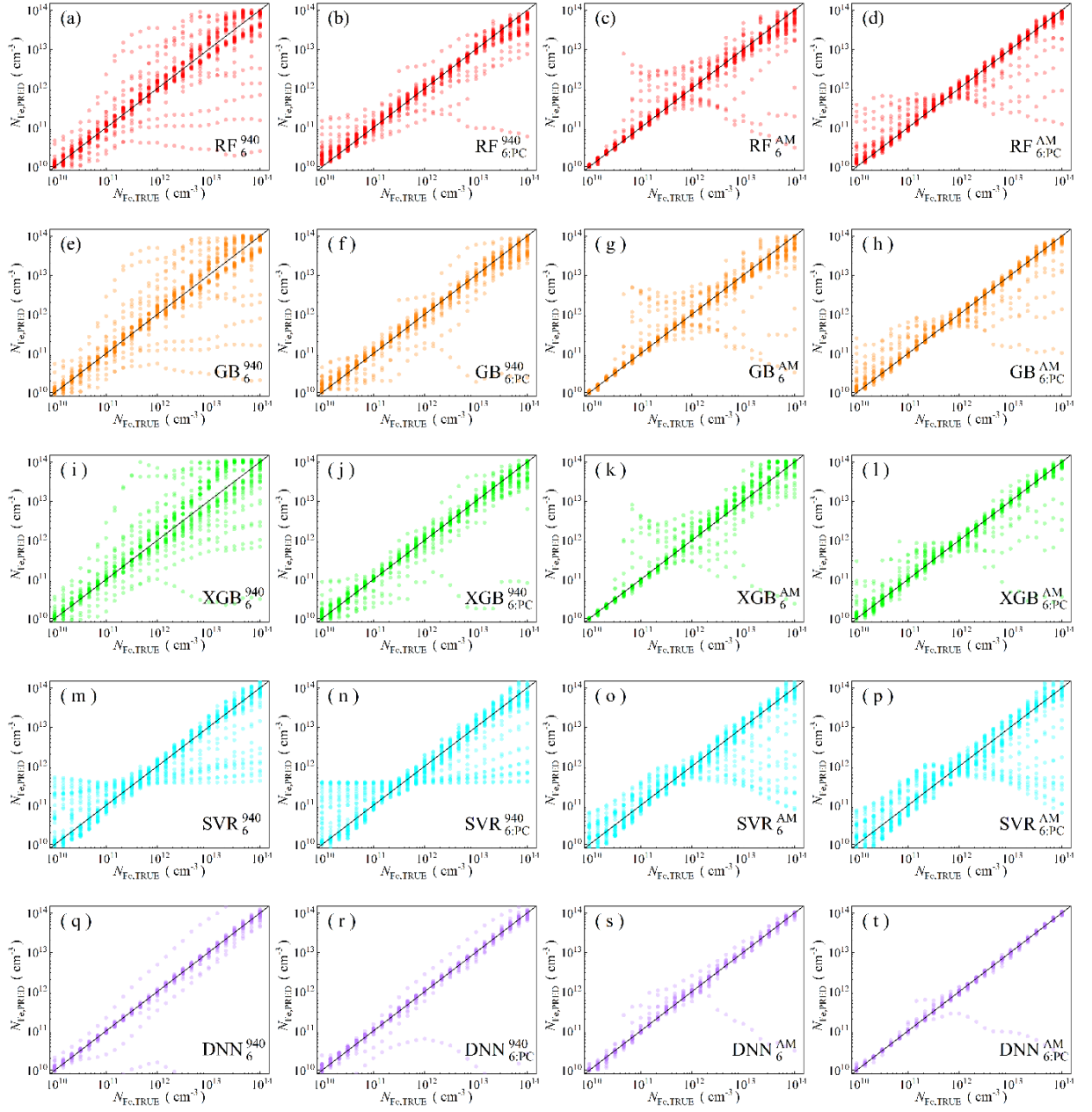


Fig.S5.18. Scatter plots compare reference iron concentrations with ML-predicted values during the N_B -altered test in the case of 6D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

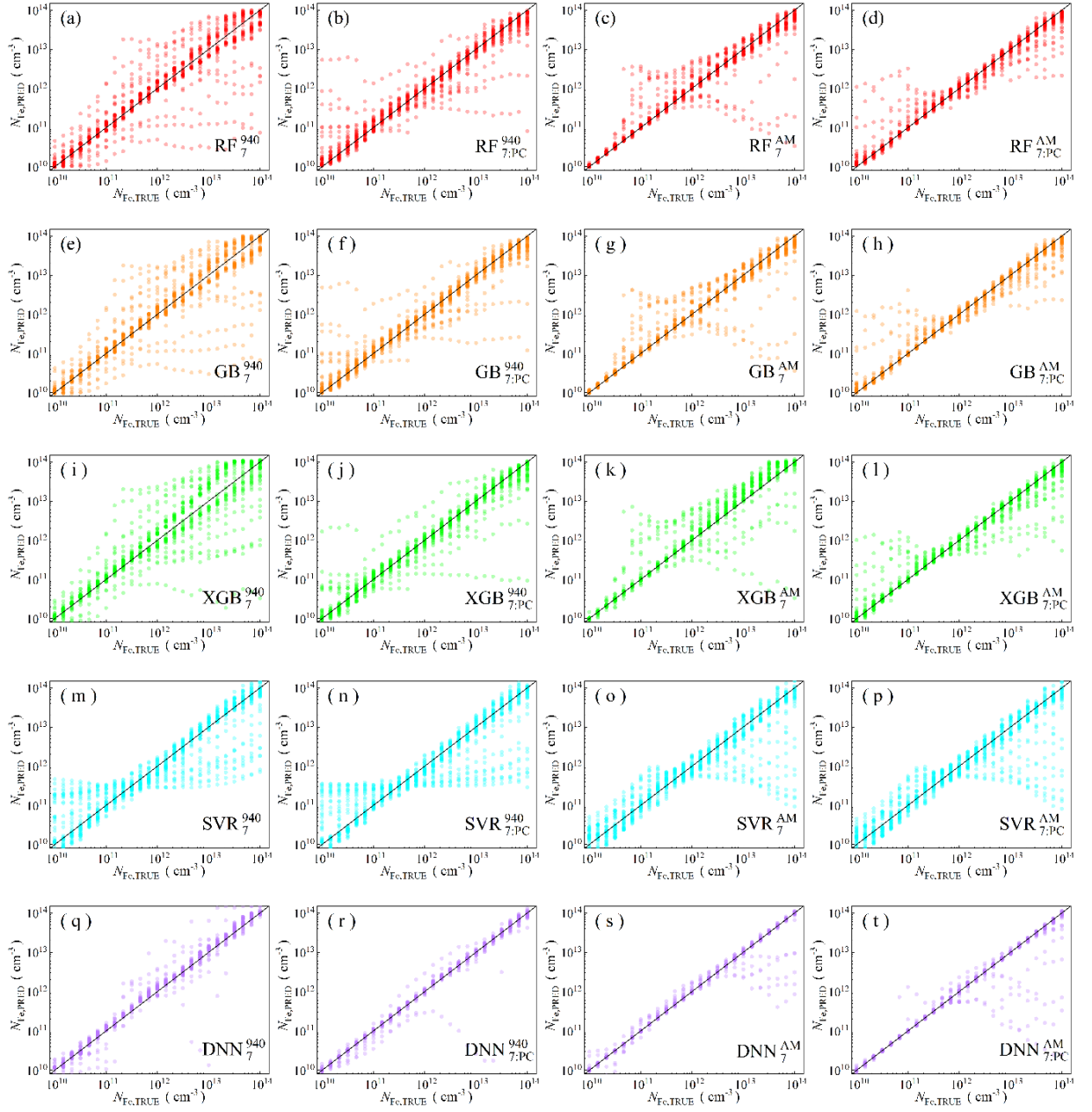


Fig.S5.19. Scatter plots compare reference iron concentrations with ML-predicted values during the N_B -altered test in the case of 7D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

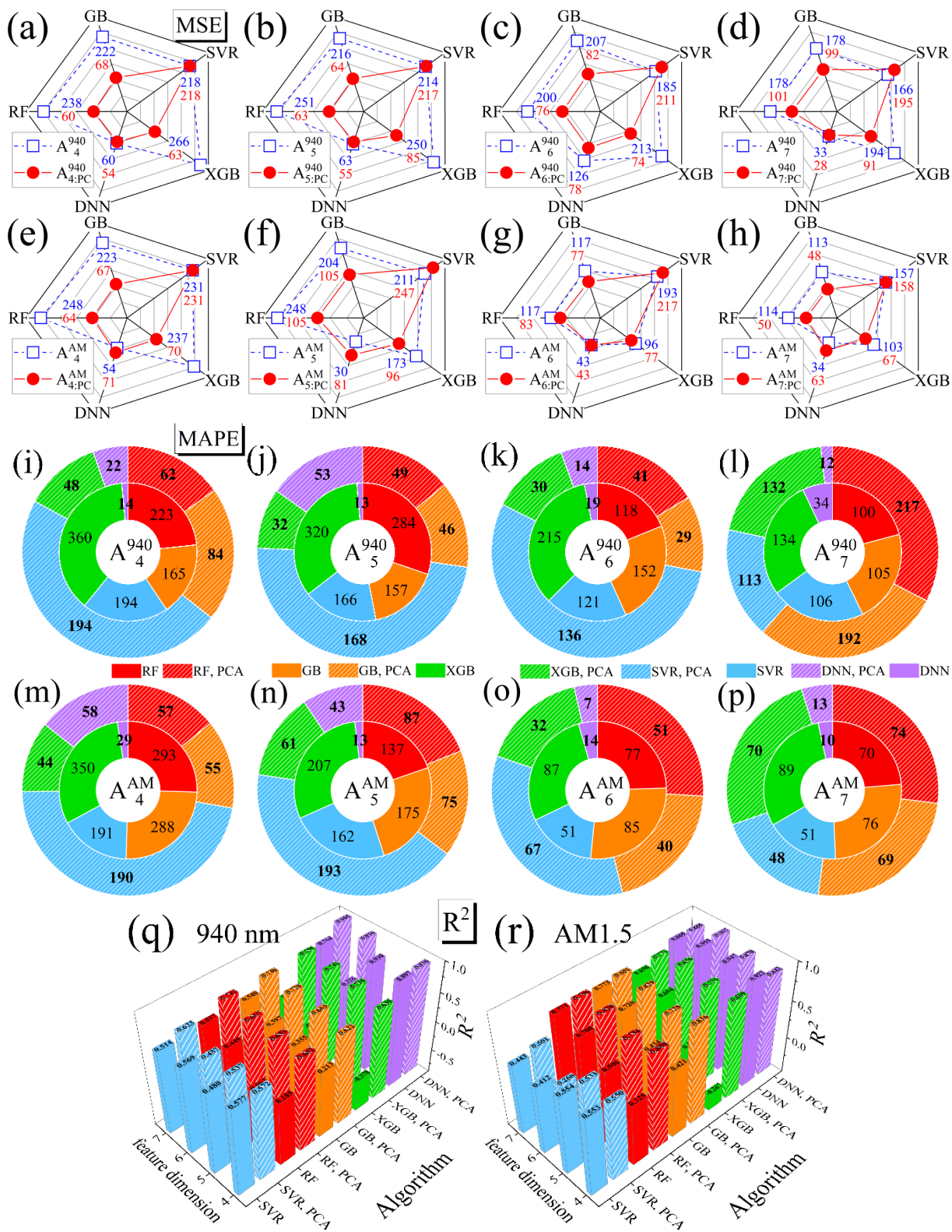


Fig.S5.20. MSE (a-h), MAPE (i-p), and R^2 (q, r) scores for various models, feature combinations, and illumination conditions on the N_B -altered test dataset. Illumination: 940 nm (a-d, i-l, q), AM1.5 (e-h, m-p, r). Feature dimensions: 4 (a, e, i, m), 5 (b, f, j, n), 6 (c, g, k, o), and 7 (d, h, l, p). Results with PCA are shown as circles (a-h) and shaded areas (i-r), while results without PCA are represented by squares (a-h) and filled regions (i-r). The numbers in panels (a-h) represent MSE values multiplied by 1000, while the numbers in panels (i-p) indicate MAPE values in percentage.

Table S5.14. Performance metrics of the models for N_B -altered dataset.

Algorithm	Feature dimension	MdAPE, %				$p01$, %				$p10$, %			
		940 nm		AM1.5		940 nm		AM1.5		940 nm		AM1.5	
		Init	PCA	Init	PCA	Init	PCA	Init	PCA	Init	PCA	Init	PCA
RF	4	46.4	19.27	44.0	16.2	0.91	5.09	1.82	4.64	11.9	28.7	16.6	34.2
	5	46.6	17.7	33.5	19.5	2.09	3.55	2.00	2.46	12.2	31.6	14.7	29.7
	6	35.7	18.4	14.4	14.4	2.18	4.27	8.46	4.64	12.7	35.1	41.1	37.9
	7	34.4	21.4	14.5	11.6	2.55	5.46	9.36	6.09	13.9	29.6	38.5	45.2
GB	4	46.1	21.3	39.7	15.7	0.91	3.55	2.82	5.55	13.6	29.3	17.4	36.7
	5	44.9	16.3	33.2	18.1	0.36	4.27	3.09	6.36	7.36	33.9	16.1	33.7
	6	34.2	13.8	11.5	12.0	1.55	7.36	14.3	4.55	18.0	42.7	47.2	43.8
	7	34.2	19.8	12.4	10.1	0.64	5.46	9.18	6.91	16.3	29.9	44.4	49.7
XGB	4	53.1	17.5	52.7	15.7	2.55	6.36	1.27	5.46	16.1	34.5	13.0	37.0
	5	53.5	15.4	40.9	14.4	3.46	5.91	2.36	5.36	15.2	38.3	14.9	40.5
	6	46.4	15.0	11.9	12.4	1.82	5.09	12.6	6.09	19.3	38.8	46.9	43.5
	7	42.0	18.4	12.2	11.4	1.36	5.36	6.91	6.91	15.8	34.0	46.6	45.7
SVR	4	32.3	31.5	36.9	35.9	1.73	1.64	1.82	1.73	15.6	17.4	16.3	16.5
	5	32.3	35.9	35.7	44.8	1.46	1.73	1.00	1.73	14.8	15.5	13.1	11.9
	6	29.6	35.3	31.6	40.7	2.64	1.09	1.73	1.46	22.5	15.9	19.4	12.4
	7	34.0	30.5	34.4	32.3	1.91	1.36	1.64	2.46	17.0	18.2	17.0	17.7
DNN	4	7.14	8.05	10.6	14.9	8.09	5.64	3.82	4.55	63.3	59.6	47.8	34.2
	5	6.05	8.06	5.76	8.36	10.7	8.00	8.64	5.91	71.4	57.4	68.0	54.9
	6	5.83	7.18	5.65	2.93	10.6	8.18	9.18	17.1	68.3	62.5	71.1	86.0
	7	17.6	6.15	2.49	3.27	1.64	10.9	23.5	16.3	27.5	67.0	82.8	81.0

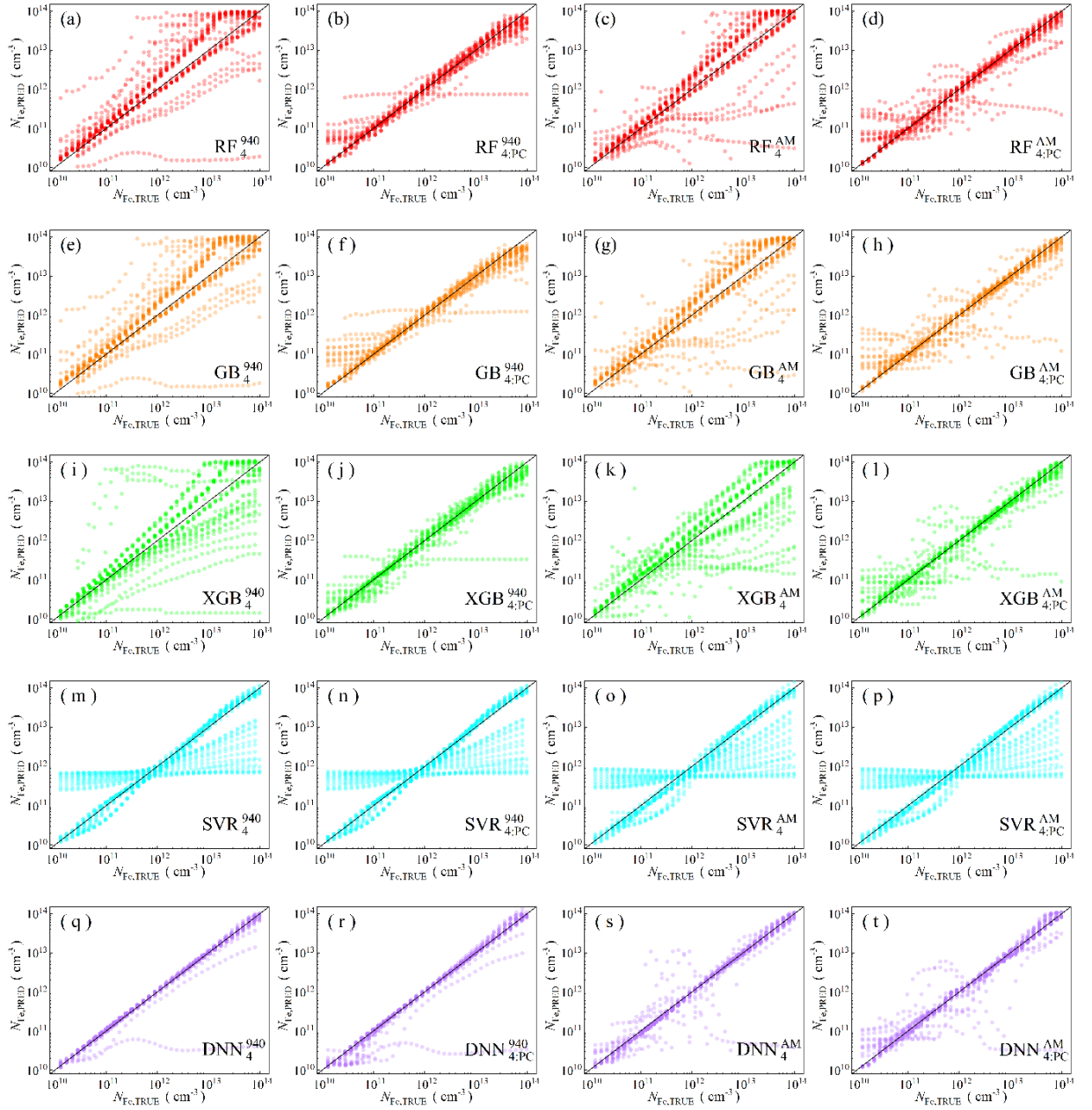


Fig.S5.21. Scatter plots compare reference iron concentrations with ML-predicted values during the All-altered test in the case of 4D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

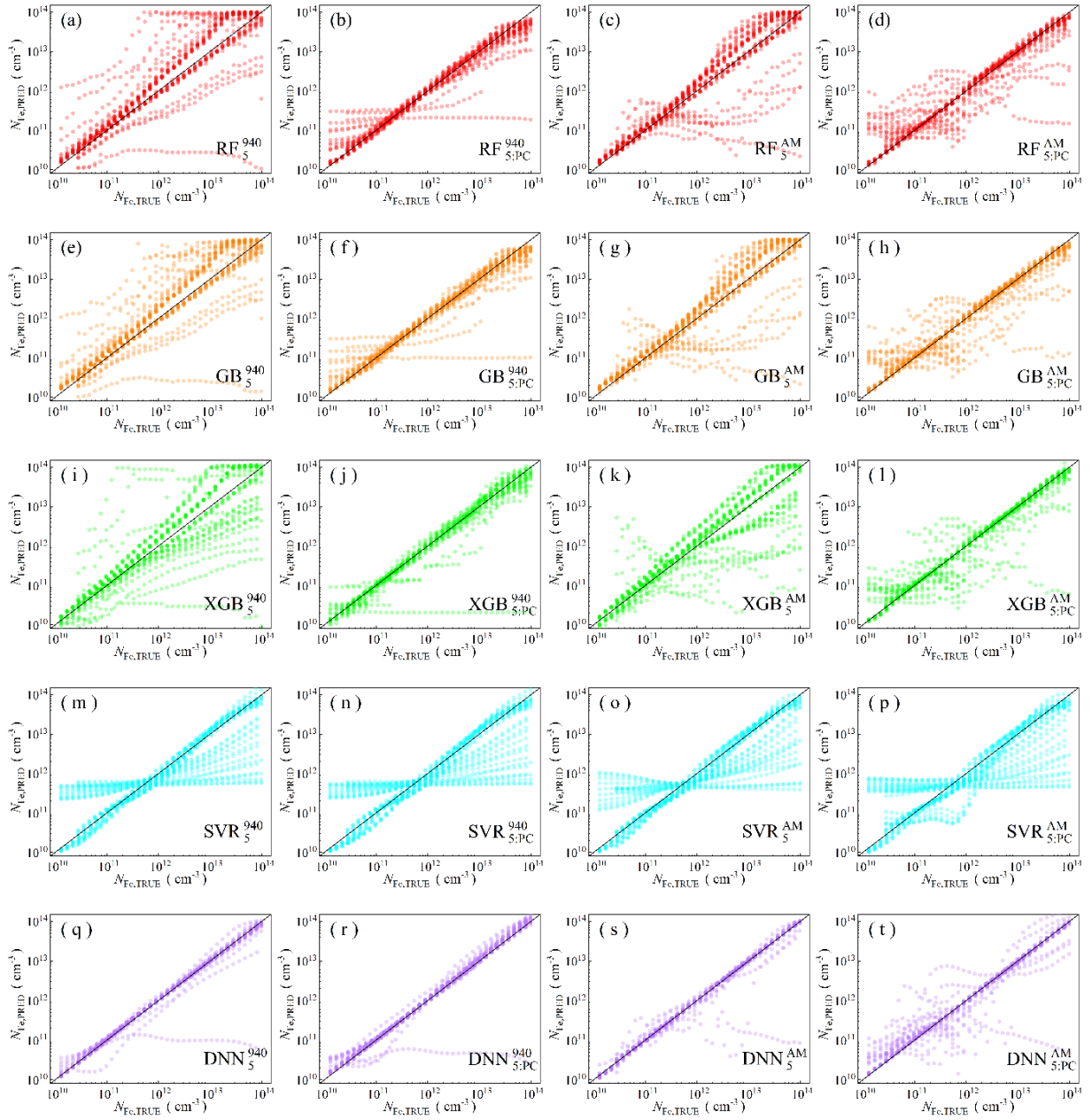


Fig.S5.22. Scatter plots compare reference iron concentrations with ML-predicted values during the All-altered test in the case of 5D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

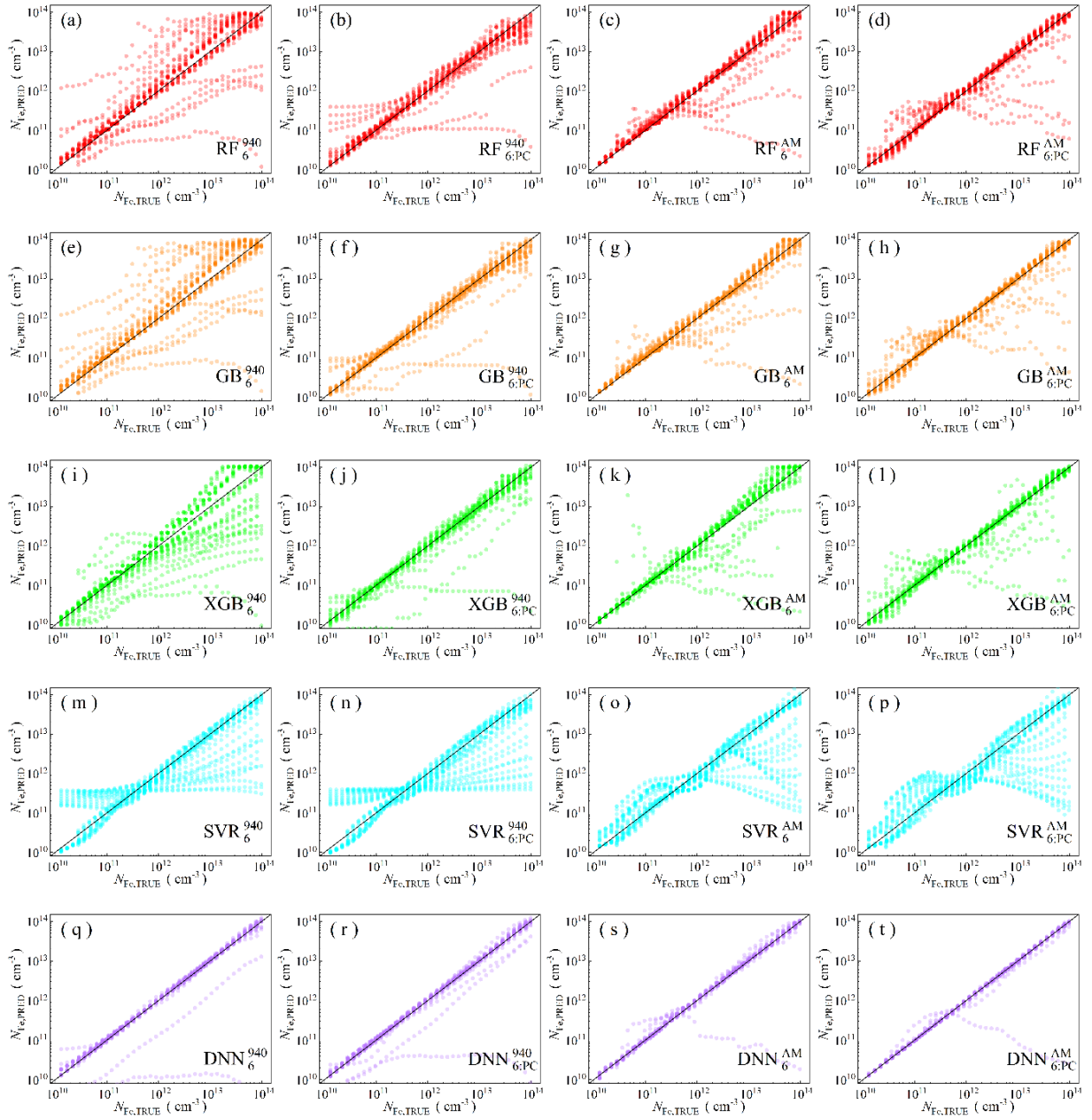


Fig.S5.23. Scatter plots compare reference iron concentrations with ML-predicted values during the All-altered test in the case of 6D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

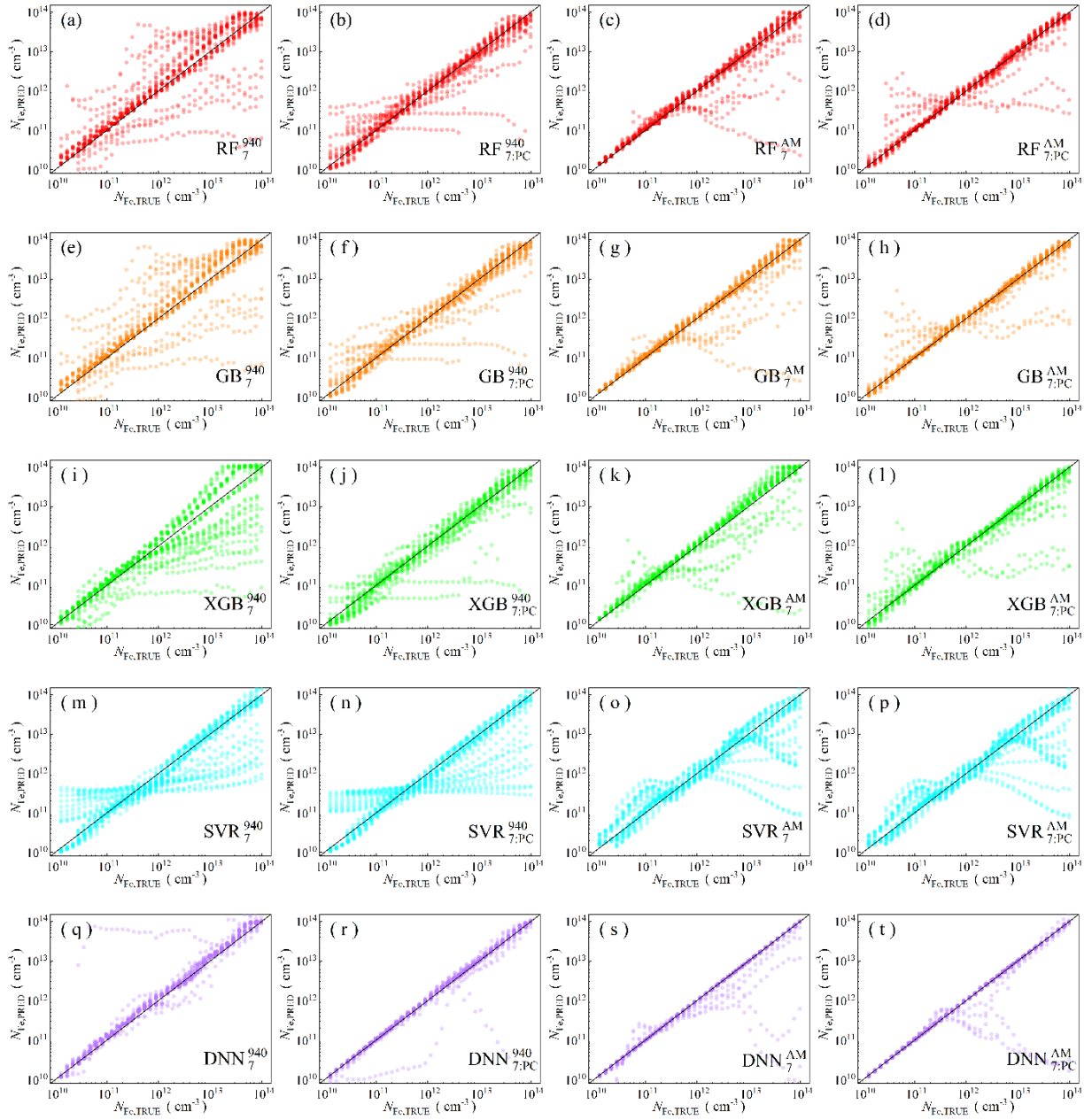


Fig.S5.24. Scatter plots compare reference iron concentrations with ML-predicted values during the All-altered test in the case of 7D features. The ML algorithms include RF (a-d), GB (e-h), XGB (i-l), SVR (m-p), and DNN (q-t). The data come from simulation under monochromatic (a, b, e, f, i, j, m, n, q, r) and AM1.5 illumination (c, d, g, h, k, l, o, p, s, t). Panels b, d, f, h, j, l, n, p, r, and t include PCA. The black lines are the identified lines serving as the references.

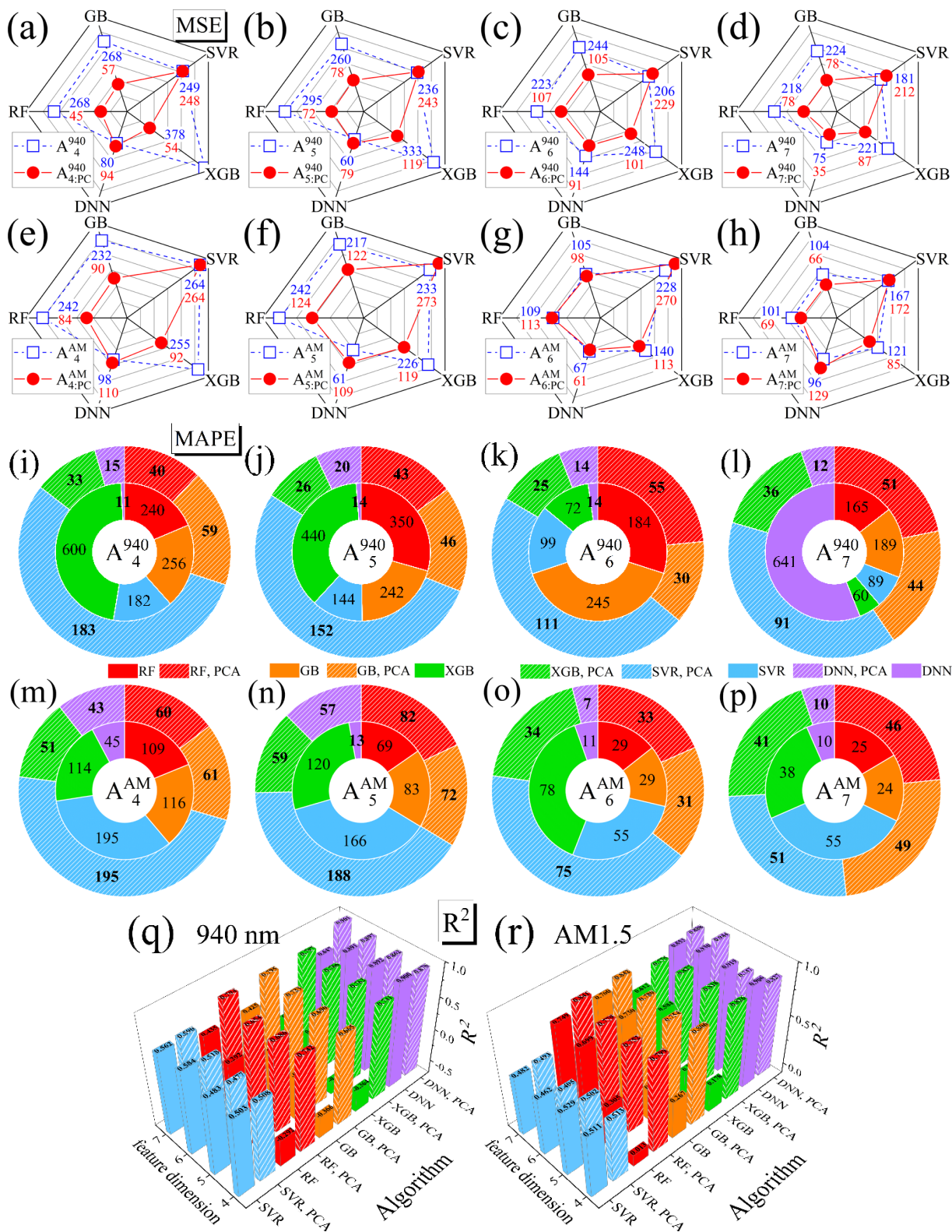


Fig.S5.25. MSE (a-h), MAPE (i-p), and R^2 (q, r) scores for various models, feature combinations, and illumination conditions on the All-altered test dataset. Illumination: 940 nm (a-d, i-l, q), AM1.5 (e-h, m-p, r). Feature dimensions: 4 (a, e, i, m), 5 (b, f, j, n), 6 (c, g, k, o), and 7 (d, h, l, p). Results with PCA are shown as circles (a-h) and shaded areas (i-r), while results without PCA are represented by squares (a-h) and filled regions (i-r). The numbers in panels (a-h) represent MSE values multiplied by 1000, while the numbers in panels (i-p) indicate MAPE values in percentage.

Table S5.15. Performance metrics of the models for All-altered dataset.

Algorithm	Feature dimension	MdAPE, %				$p01$, %				$p10$, %			
		940 nm		AM1.5		940 nm		AM1.5		940 nm		AM1.5	
		Init	PCA	Init	PCA	Init	PCA	Init	PCA	Init	PCA	Init	PCA
RF	4	55.8	17.2	42.8	16.0	0.92	4.37	1.93	4.03	10.7	35.5	13.2	35.8
	5	55.2	17.4	33.4	18.2	1.26	3.53	1.35	3.70	10.3	32.4	15.8	34.6
	6	41.6	19.3	15.6	14.5	2.02	3.95	2.52	4.37	17.8	32.7	33.5	39.3
	7	38.4	20.4	14.3	11.4	1.01	3.03	3.61	5.21	15.5	29.2	36.2	44.5
GB	4	58.4	17.5	53.5	15.1	1.01	4.45	1.18	4.45	10.8	33.3	12.4	36.1
	5	56.1	16.5	39.4	15.6	0.76	4.54	1.60	4.29	11.7	37.0	15.3	38.1
	6	43.1	13.4	14.5	12.7	1.85	5.13	4.03	6.05	16.7	41.5	36.0	43.3
	7	39.9	19.9	13.3	10.4	0.92	3.70	2.69	6.47	15.0	30.8	40.6	49.1
XGB	4	57.6	18.0	65.1	14.8	0.42	5.46	0.84	4.20	7.48	34.7	7.90	38.8
	5	59.3	14.8	49.7	12.1	1.09	4.37	1.26	5.71	8.91	39.6	9.92	44.5
	6	45.7	15.0	16.8	12.3	0.42	6.05	3.11	3.87	11.2	38.7	33.8	43.8
	7	44.5	22.7	20.3	11.7	1.35	2.69	1.77	5.29	8.24	30.3	27.7	45.5
SVR	4	30.6	28.4	34.5	33.8	1.77	1.43	1.68	2.02	17.7	19.6	15.1	16.1
	5	32.8	37.9	35.1	41.7	1.68	1.77	1.68	1.01	15.9	14.0	15.4	13.1
	6	29.8	37.3	31.0	41.0	2.27	1.26	1.85	1.68	20.9	13.3	18.2	12.4
	7	34.0	30.7	32.5	29.0	2.02	2.10	1.68	2.35	17.1	17.0	15.2	18.7
DNN	4	6.29	7.34	11.6	13.4	8.07	6.98	3.36	3.53	67.1	63.0	43.4	39.9
	5	6.95	10.2	5.51	7.74	7.65	4.62	10.3	7.56	64.5	49.4	71.7	58.0
	6	6.52	7.60	5.13	3.06	8.24	9.33	11.0	17.7	69.4	60.5	73.4	88.0
	7	19.1	7.16	2.05	3.46	2.69	8.99	27.8	17.1	28.6	63.3	85.4	80.6

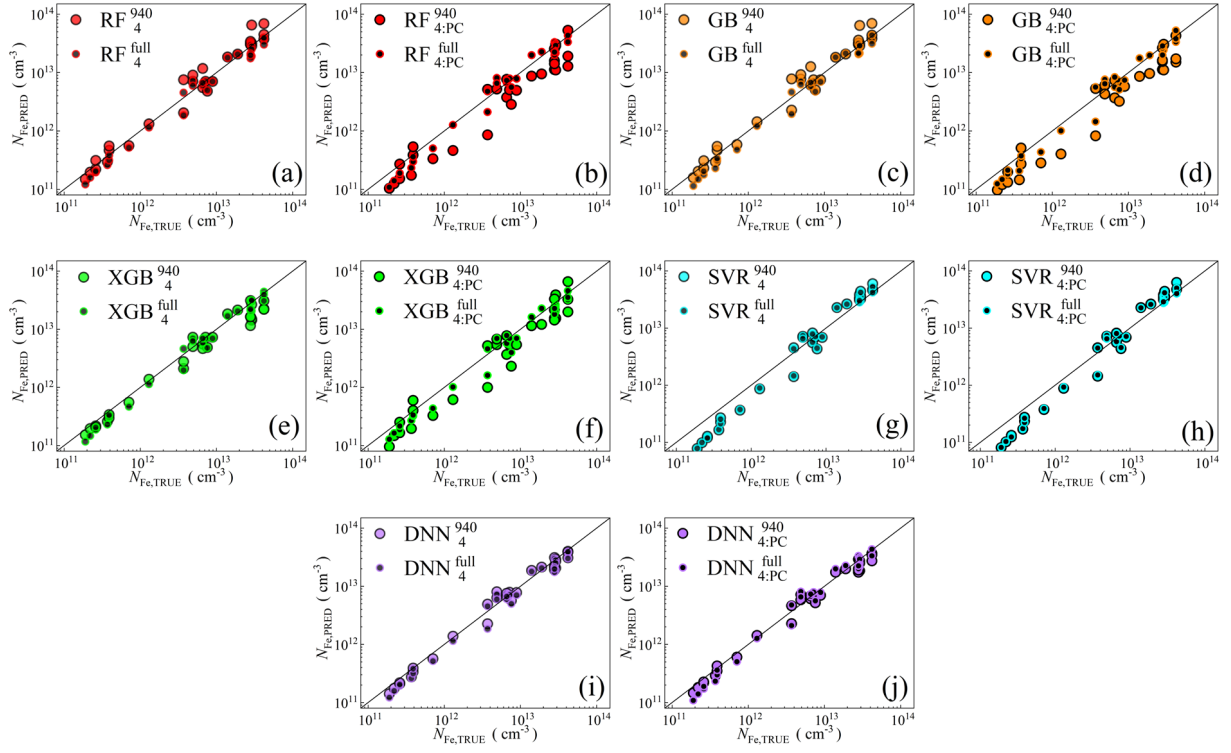


Fig.S5.26. Scatter plots compare reference iron concentrations with ML-predicted values for experimental dataset in the case of 4D features. The ML algorithms include RF (a, b), GB (c, d), XGB (e, f), SVR (g, h), and DNN (i, j). Panels b, d, f, h, and j include PCA. The black lines are the identified lines serving as the references.

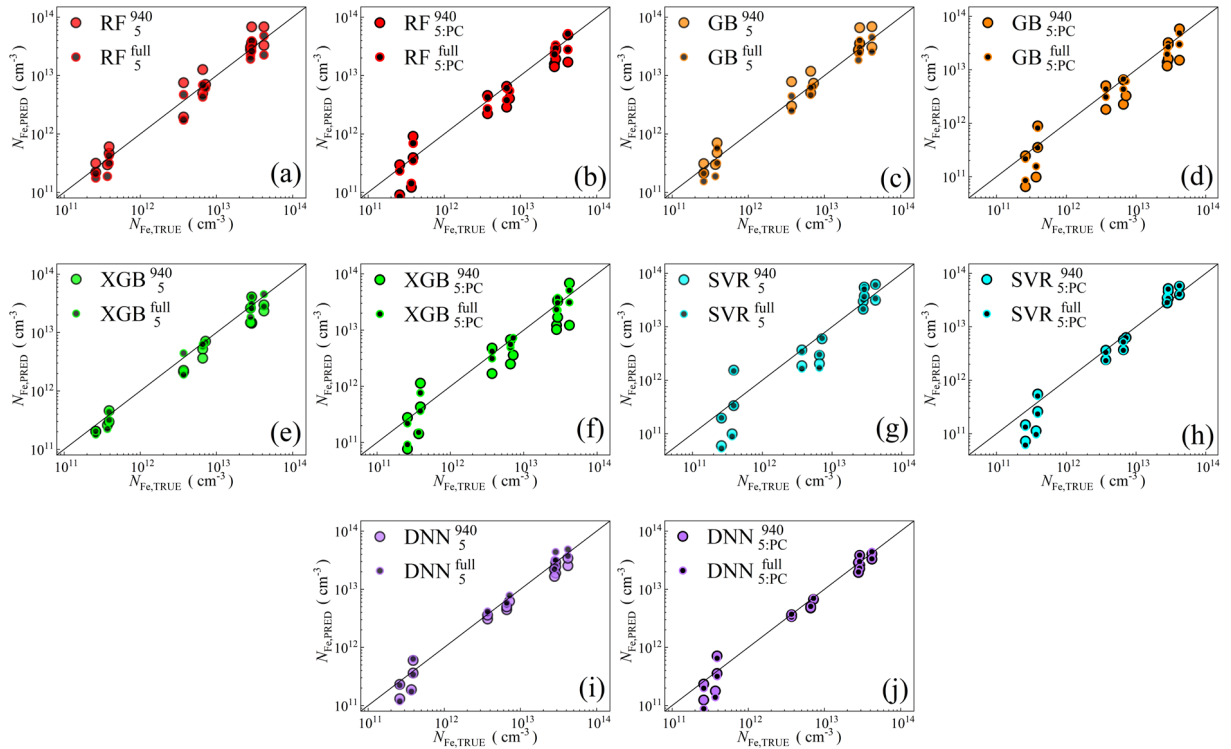


Fig.S5.27. Scatter plots compare reference iron concentrations with ML-predicted values for experimental dataset in the case of 5D features. The ML algorithms include RF (a, b), GB (c, d), XGB (e, f), SVR (g, h), and DNN (i, j). Panels b, d, f, h, and j include PCA. The black lines are the identified lines serving as the references.

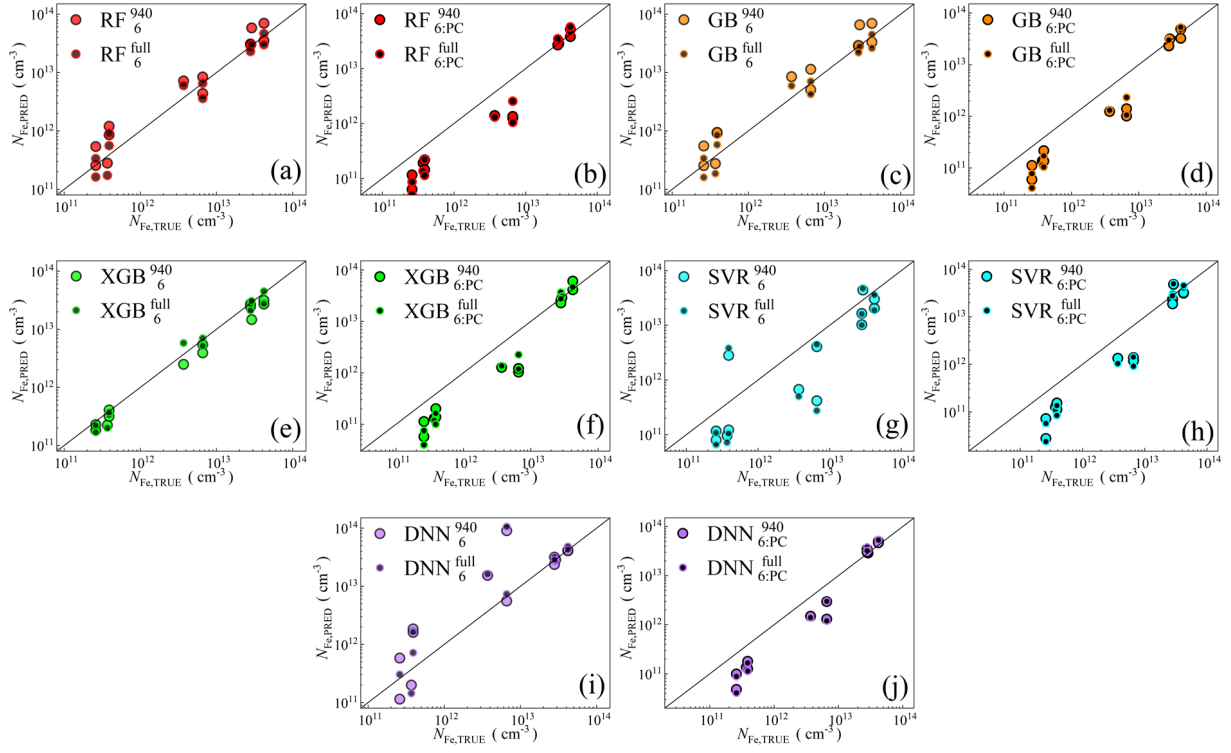


Fig.S5.28. Scatter plots compare reference iron concentrations with ML-predicted values for experimental dataset in the case of 6D features. The ML algorithms include RF (a, b), GB (c, d), XGB (e, f), SVR (g, h), and DNN (i, j). Panels b, d, f, h, and j include PCA. The black lines are the identified lines serving as the references.

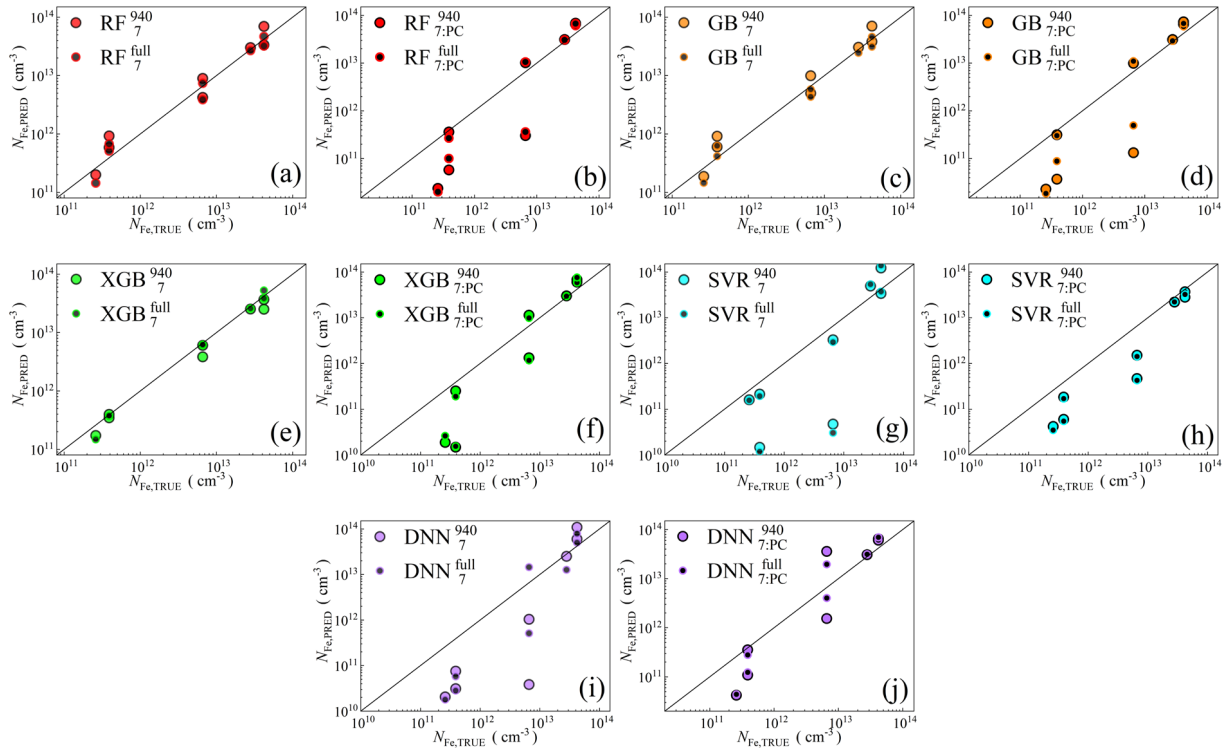


Fig.S5.29. Scatter plots compare reference iron concentrations with ML-predicted values for experimental dataset in the case of 7D features. The ML algorithms include RF (a, b), GB (c, d), XGB (e, f), SVR (g, h), and DNN (i, j). Panels b, d, f, h, and j include PCA. The black lines are the identified lines serving as the references.

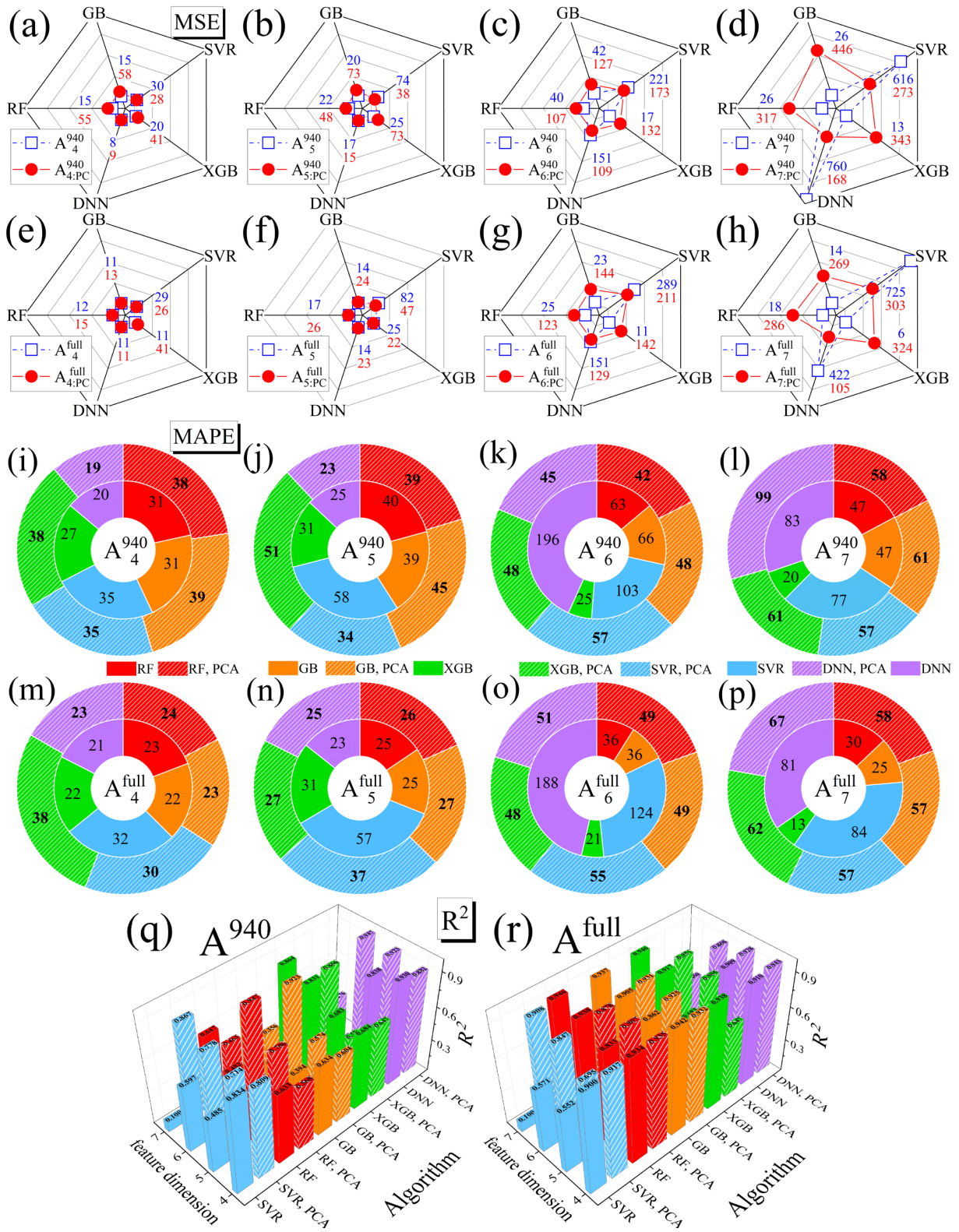


Fig.S5.30. MSE (a-h), MAPE (i-p), and R^2 (q, r) scores for various models and feature combinations on the experimental test dataset. The models were trained either on the train dataset (a-d, i-l, q) or on the entire set of synthetic data (e-h, m-p, r).. Feature dimensions: 4 (a, e, i, m), 5 (b, f, j, n), 6 (c, g, k, o), and 7 (d, h, l, p). Results with PCA are shown as circles (a-h) and shaded areas (i-r), while results without PCA are represented by squares (a-h) and filled regions (i-r). The numbers in panels (a-h) represent MSE values multiplied by 1000, while the numbers in panels (i-p) indicate MAPE values in percentage. The number of samples in the experimental test set was 30, 17, 13, and 8 for models with feature dimensions of 4, 5, 6, and 7, respectively.

Table S5.14. Performance metrics of the models for N_{Fe} -altered dataset.

Algorithm	Feature dimension	MdAPE, %				$p01$, %				$p10$, %			
		940 nm		AM1.5		940 nm		AM1.5		940 nm		AM1.5	
		Init	PCA	Init	PCA	Init	PCA	Init	PCA	Init	PCA	Init	PCA
RF	4	6.47	6.38	8.29	8.00	10.1	9.48	7.54	8.12	68.0	70.2	57.0	57.9
	5	7.67	6.00	6.96	9.46	7.06	8.51	10.1	6.96	60.5	72.2	64.2	52.1
	6	8.01	7.34	5.46	5.66	6.58	7.74	11.0	10.9	59.5	60.4	72.4	72.9
	7	8.89	8.74	6.27	4.62	5.71	5.42	10.4	13.3	54.2	56.4	68.8	82.2
GB	4	4.63	4.35	6.95	6.27	13.6	14.1	9.87	9.38	76.5	80.3	62.4	67.2
	5	4.96	4.63	6.55	8.05	13.6	12.5	11.2	7.25	72.7	82.4	66.6	57.4
	6	6.78	6.73	5.31	4.41	8.03	9.09	11.9	15.4	65.5	64.0	74.4	79.1
	7	7.87	7.41	4.97	3.73	7.16	7.64	12.2	17.4	60.7	61.8	75.3	84.3
XGB	4	4.82	6.91	6.43	7.75	11.0	9.77	10.1	8.32	75.6	63.5	62.7	60.2
	5	4.46	6.20	4.26	9.04	11.7	10.2	13.8	6.58	78.6	68.6	74.7	52.3
	6	6.49	7.57	4.13	4.79	9.87	8.51	14.8	11.8	68.7	59.9	83.2	77.2
	7	6.16	7.96	3.71	3.94	8.80	7.54	15.2	15.6	66.0	60.9	83.3	83.0
SVR	4	31.1	26.7	41.2	40.5	1.45	1.35	1.06	0.77	14.9	18.2	11.4	11.8
	5	36.3	42.7	39.7	51.9	1.64	0.87	1.74	1.35	14.4	10.8	10.9	9.77
	6	37.0	44.8	37.3	44.2	1.16	0.48	1.64	0.77	13.4	8.99	14.4	10.1
	7	39.1	38.1	36.5	32.5	1.16	1.06	0.68	1.26	9.48	13.7	13.4	15.2
DNN	4	8.06	7.48	11.4	13.0	7.74	7.64	3.97	4.35	58.8	60.5	44.8	41.6
	5	7.42	9.02	5.97	6.84	6.48	6.67	10.1	6.77	66.0	54.7	72.3	61.4
	6	6.98	8.15	6.30	3.18	7.06	5.80	8.80	17.8	64.8	59.3	69.9	90.2
	7	17.3	6.93	1.63	2.40	4.06	8.90	32.6	23.5	33.1	64.7	98.7	96.7

Table S5.15. Performance metrics of the models for *T*-altered dataset.

Algorithm	Feature dimension	MdAPE, %				<i>p</i> 01, %				<i>p</i> 10, %			
		940 nm		AM1.5		940 nm		AM1.5		940 nm		AM1.5	
		Init	PCA	Init	PCA	Init	PCA	Init	PCA	Init	PCA	Init	PCA
RF	4	6.50	13.1	1.88	11.6	18.2	8.92	28.1	11.8	60.4	40.8	95.6	45.2
	5	5.59	13.2	9.58	5.90	22.4	6.75	12.1	16.6	62.9	40.7	51.3	62.7
	6	4.61	13.3	14.2	2.13	20.8	8.58	9.50	35.3	68.8	43.5	42.1	77.3
	7	5.07	9.82	5.05	1.94	21.5	12	17.7	38.7	67.4	50.4	68.3	79.8
GB	4	5.29	12.5	3.47	10.1	20.7	6.67	21.5	5.08	62.0	42.9	80.6	49.8
	5	4.91	12.6	9.34	4.89	19.1	7.00	6.58	22.5	66.2	44.7	51.8	65.0
	6	3.41	9.12	12.8	1.81	22.3	11.2	4.67	40.9	73.1	52.4	43.2	78.3
	7	5.03	9.72	5.14	2.44	15.6	10.2	13.9	31.3	68.3	51.0	70.7	79.3
XGB	4	5.03	10.9	3.40	8.72	22.8	8.92	20.8	8.42	64.6	46.9	80.8	53.8
	5	3.64	9.01	9.88	5.14	28.4	8.75	7.00	23.7	67.3	53.0	50.3	66.3
	6	2.47	10.4	10.5	2.08	33.6	6.75	6.58	34.4	73.7	49.2	48.6	78.0
	7	9.93	5.37	5.78	3.38	9.92	14.9	12.3	19.3	50.3	67.9	69.2	78.3
SVR	4	30.9	27.8	5.11	36.4	1.50	1.58	13.6	0.67	15.3	18.3	72.9	12.8
	5	33.7	38.0	35.7	37.5	1.75	1.00	1.17	0.92	15.3	12.9	13.5	12.8
	6	38.3	40.9	47.6	37.1	1.50	1.50	0.67	1.25	14.3	15.0	9.83	12.8
	7	18.6	6.85	46.2	36.3	3.83	7.50	0.75	1.83	29.8	66.3	9.58	15.4
DNN	4	6.92	7.54	10.5	14.2	7.83	8.33	4.58	4.67	65.3	59.0	47.3	37.2
	5	6.53	8.90	5.46	7.44	9.17	5.50	10.9	7.42	68.8	53.8	71.6	58.4
	6	7.09	7.18	4.92	2.99	7.50	9.25	10.7	18.3	66.8	61.1	79.2	90.9
	7	39.3	35.9	32.2	2.53	1.25	1.50	1.00	23.2	10.7	13.3	14.7	92.5